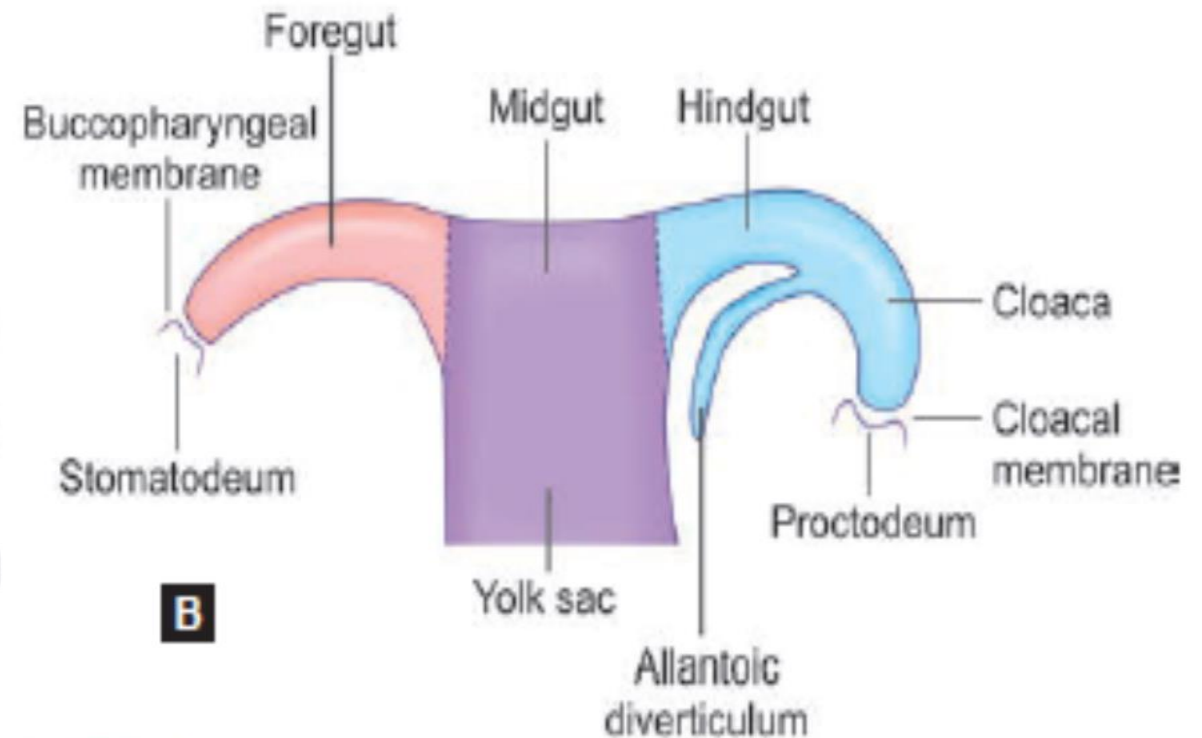
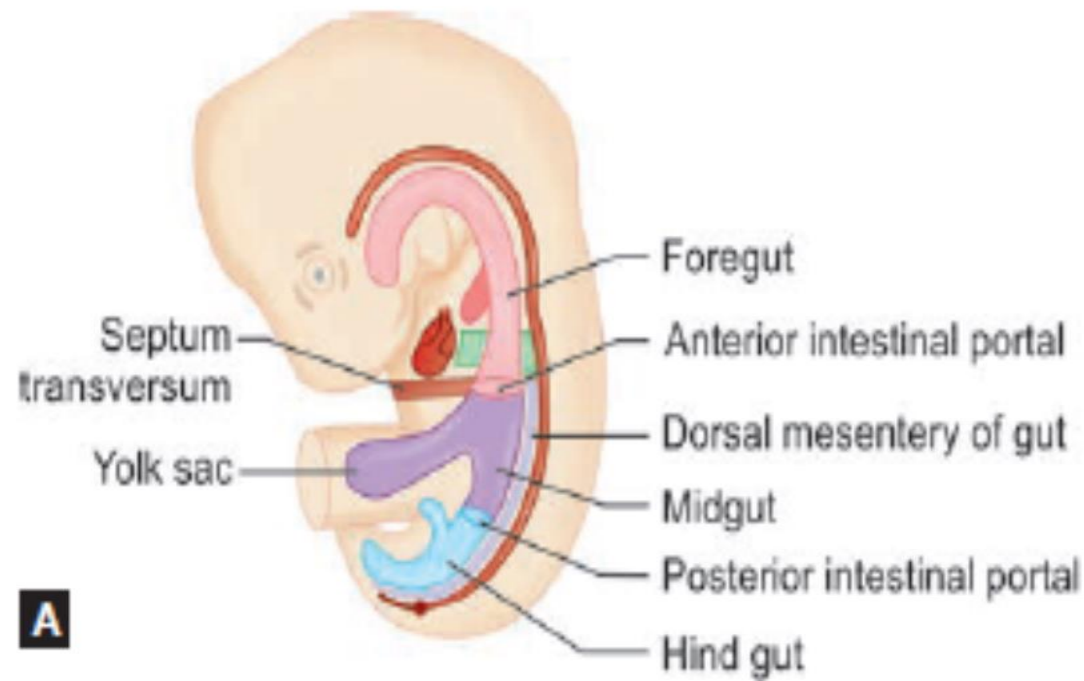
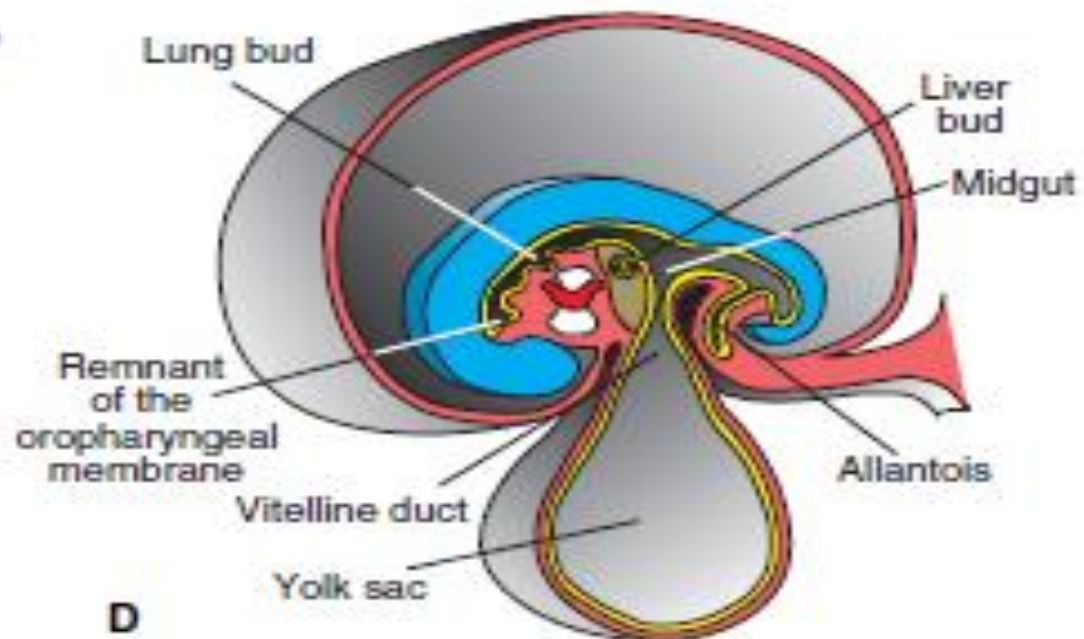
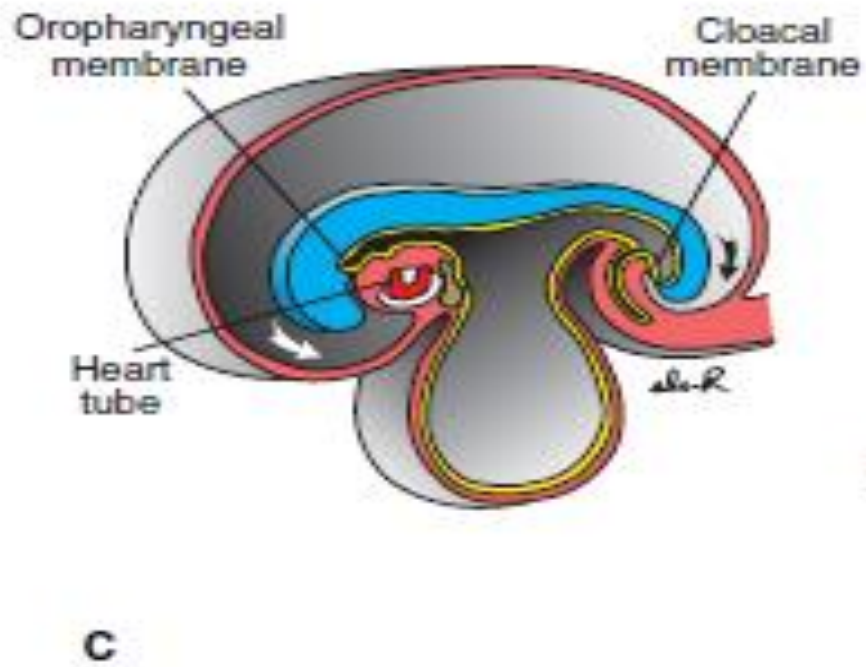
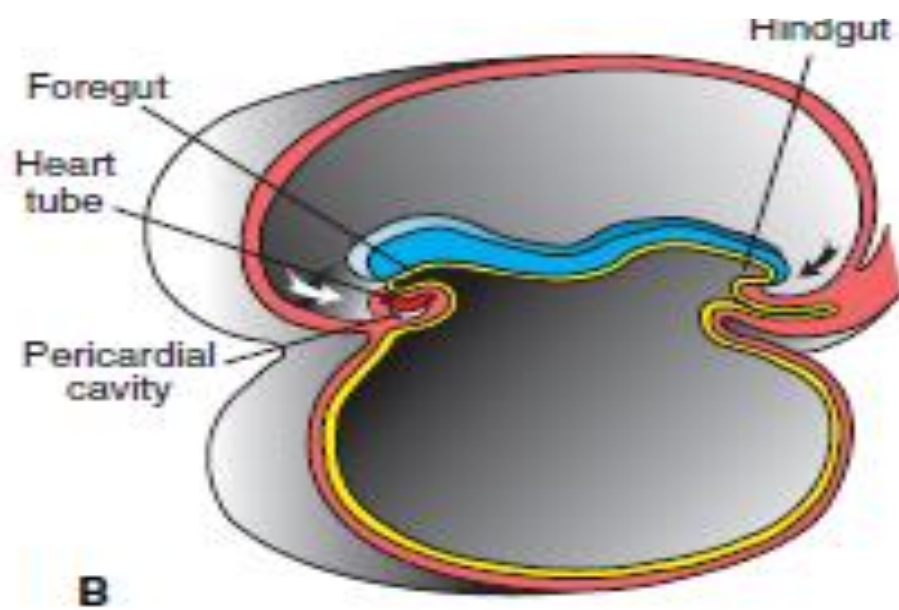
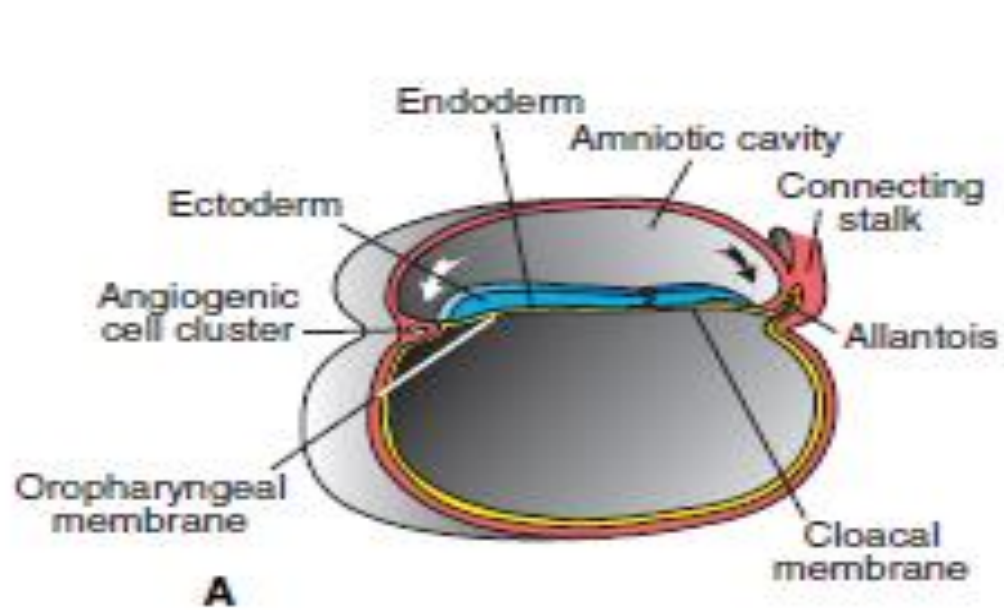


The Digestive System

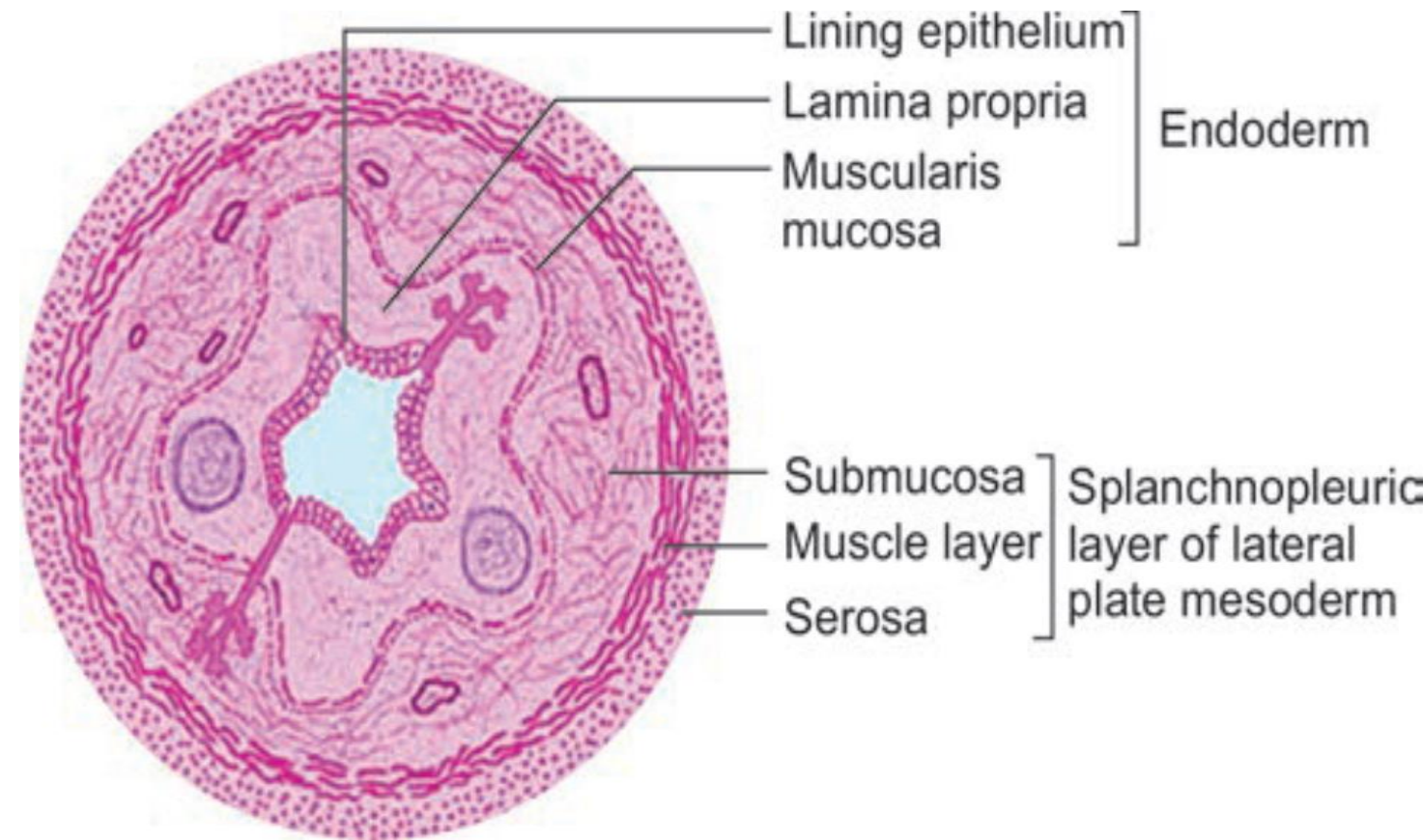
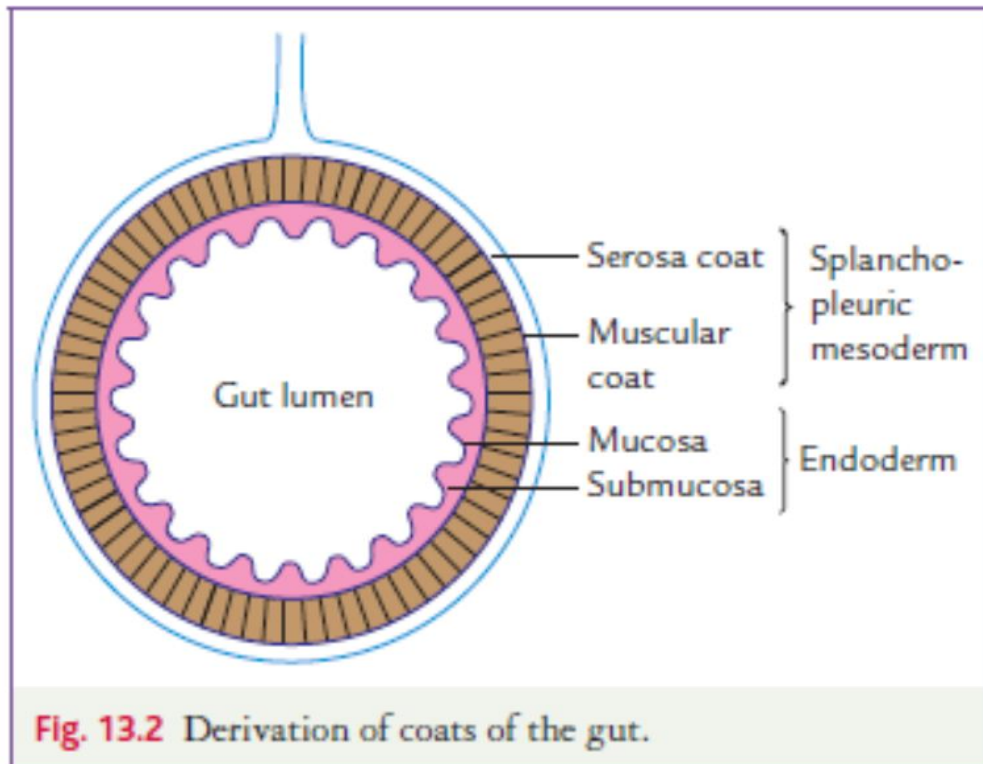
D.CHIKWANDA

- Cephalocaudal & lateral folding of the embryo, causes a dorsal portion of the endoderm-lined yolk sac cavity incorporation into the embryo to form the **primitive gut in the 4th week**
- **primordial gut** at the beginning of the 4th week is closed at its cranial end by the **oropharyngeal membrane** and at its caudal end by the **cloacal membrane**
- 2 other portions of the endoderm-lined cavity, the **yolk sac & the allantois**, remain outside the embryo

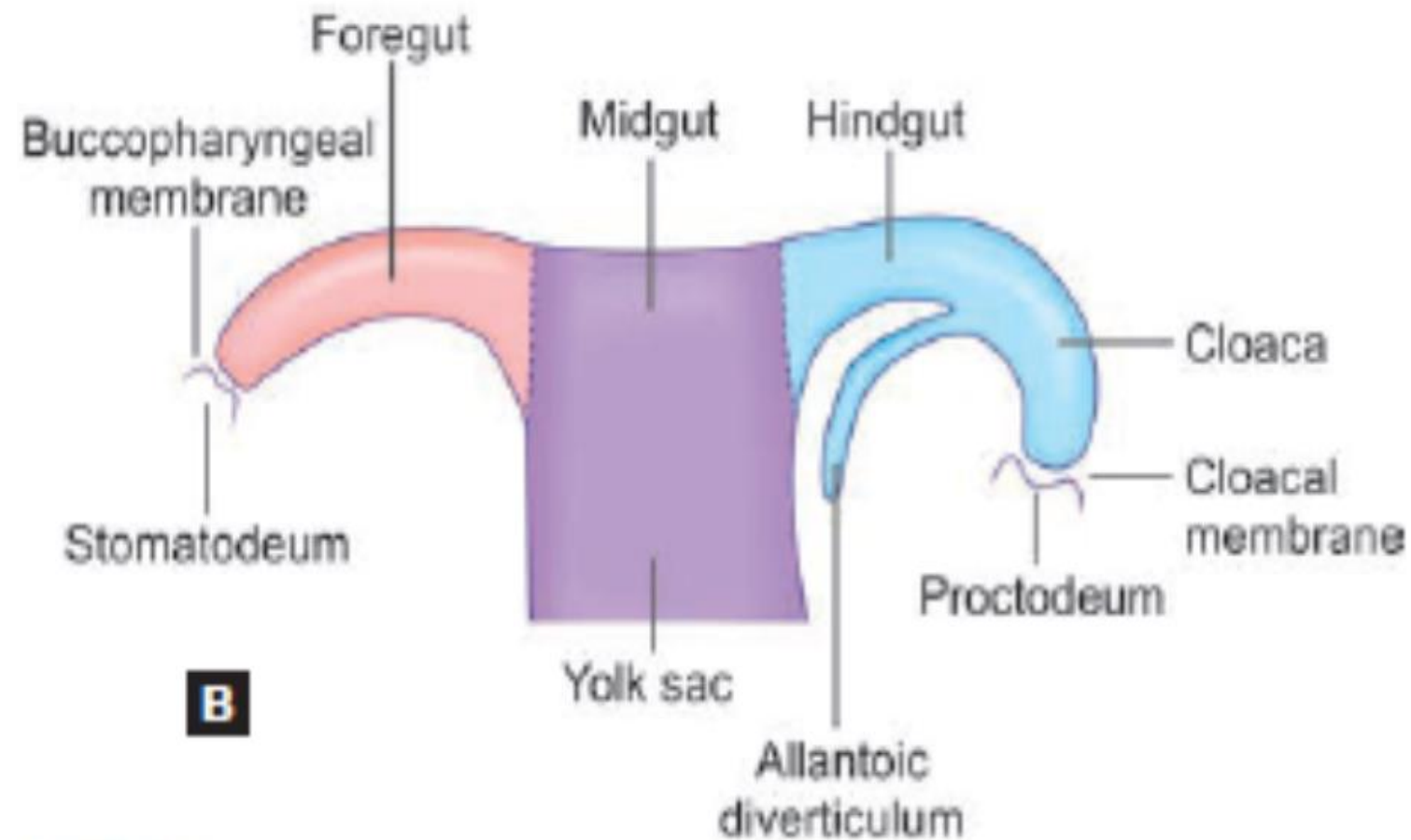
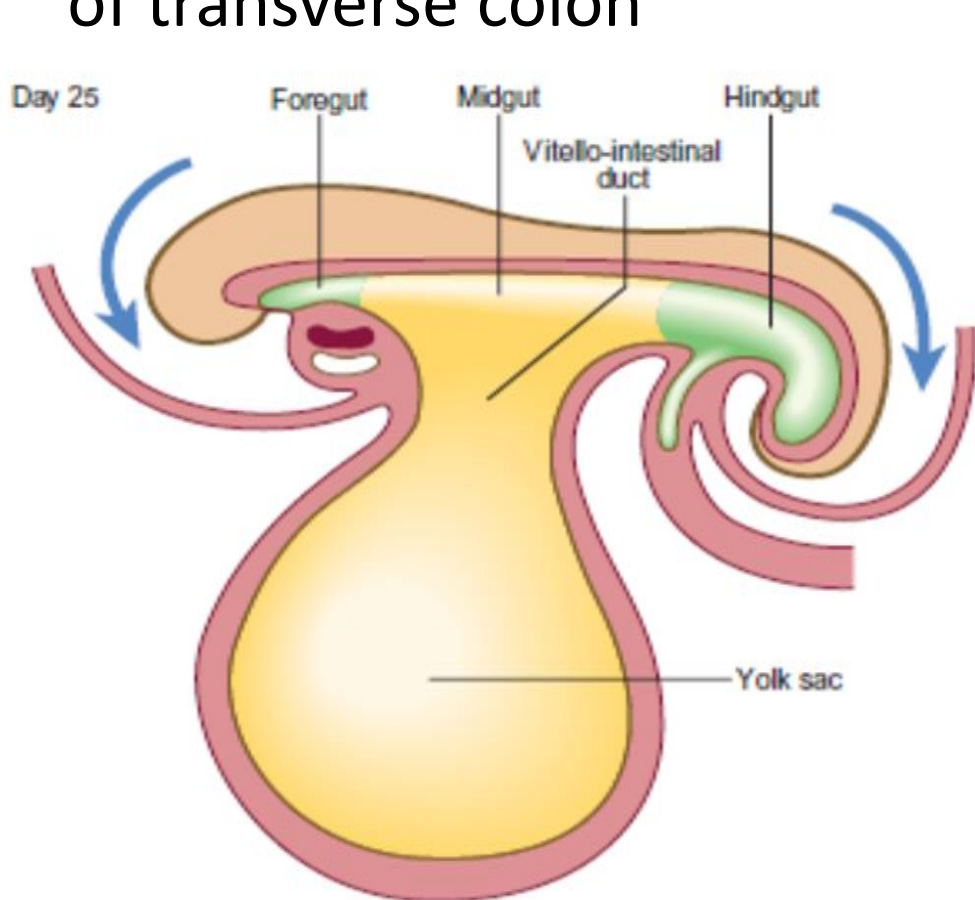




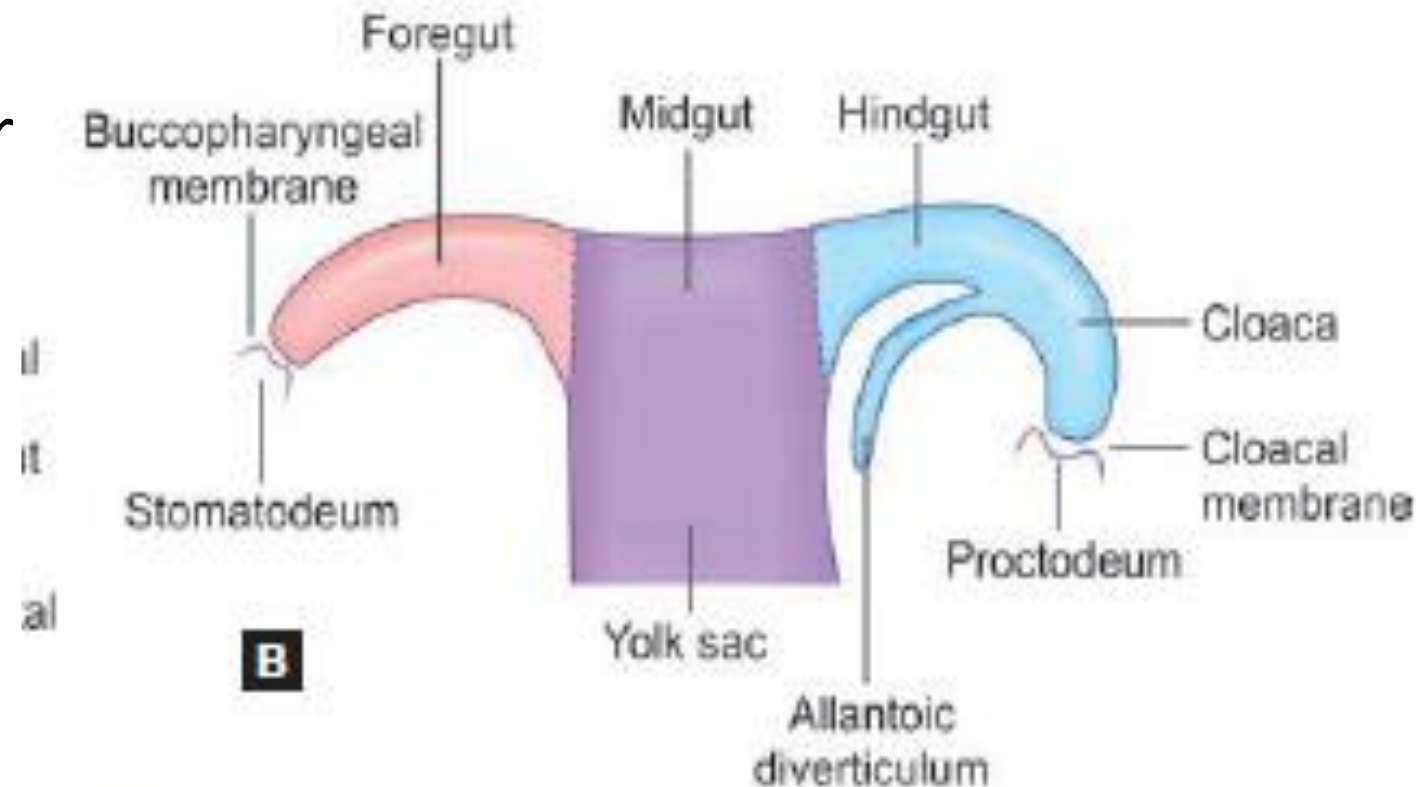
- endoderm of the primordial gut gives rise to most of its epithelium & glands
- muscular, connective tissue, & other layers of the wall of the digestive tract are derived from splanchnic mesenchyme surrounding the primordial gut
- primordial gut is divided into 3: **foregut, midgut & hindgut**
- *Hox* and *ParaHox* genes, as well as Shh signals, regulate the regional differentiation of the primordial gut to form the different parts



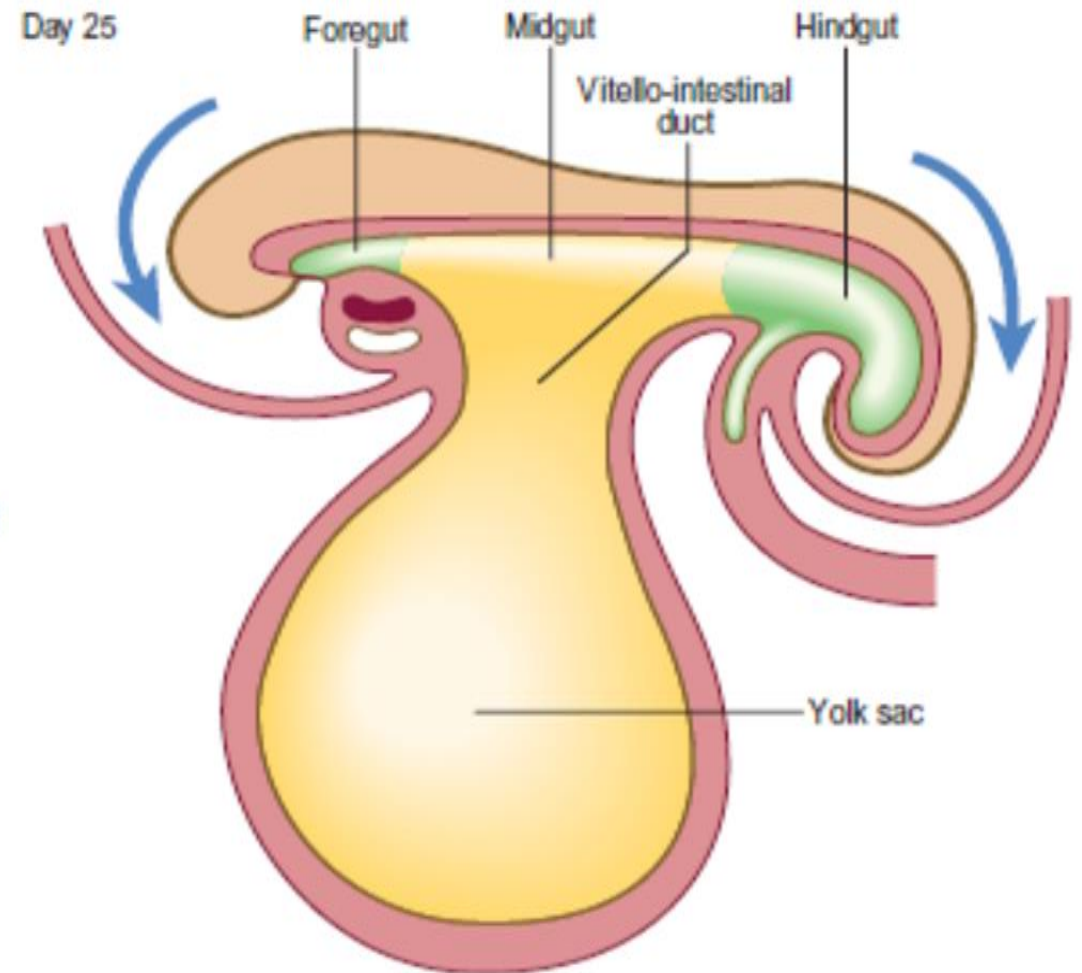
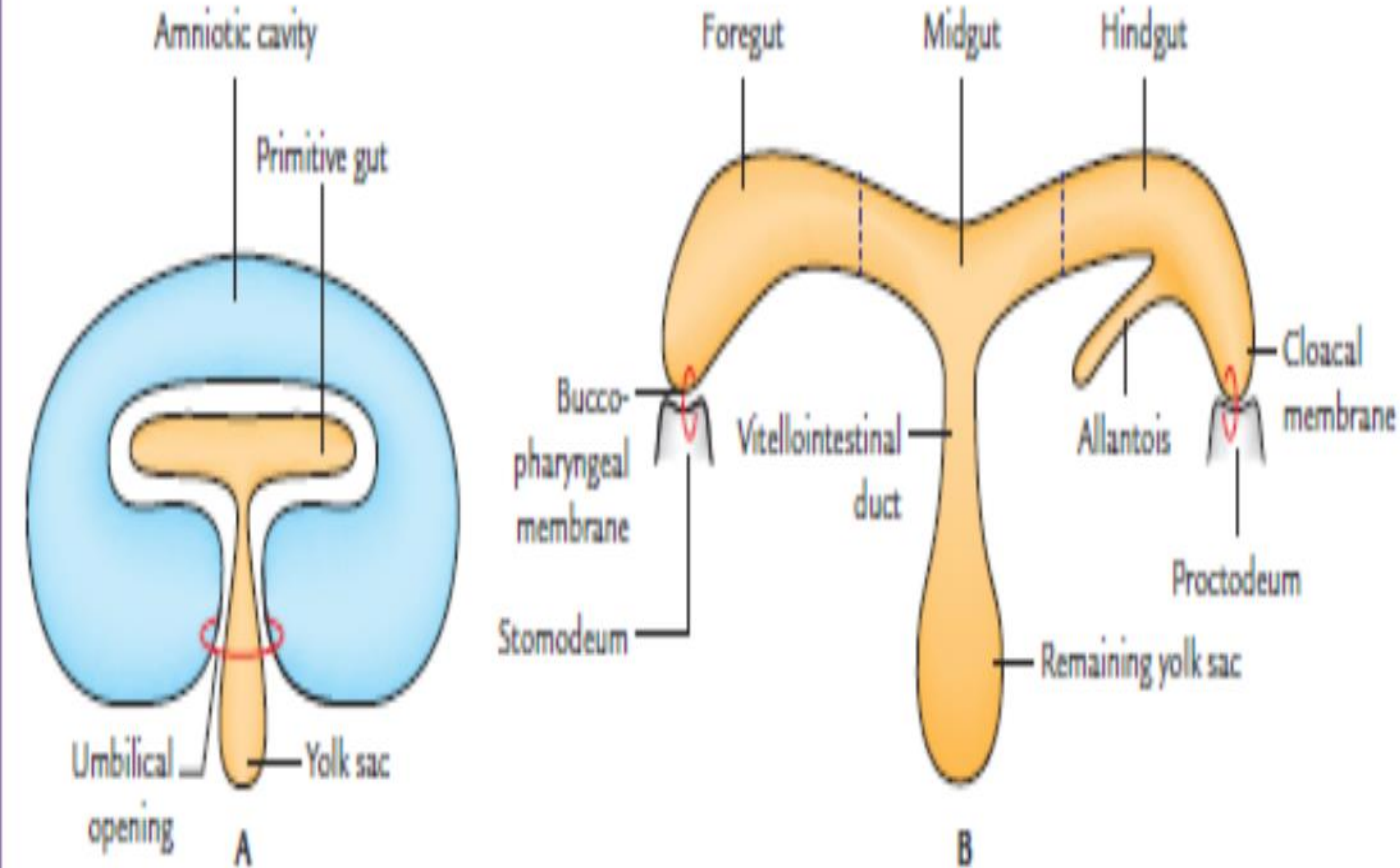
- Communication between foregut & midgut is - ***anterior intestinal portal***
- represented in an adult by the termination of bile duct in the 2nd part of duodenum
- communication between midgut & hindgut is - ***posterior intestinal portal***
- corresponds in the adult to the junction of right two-thirds with the left one-third of transverse colon



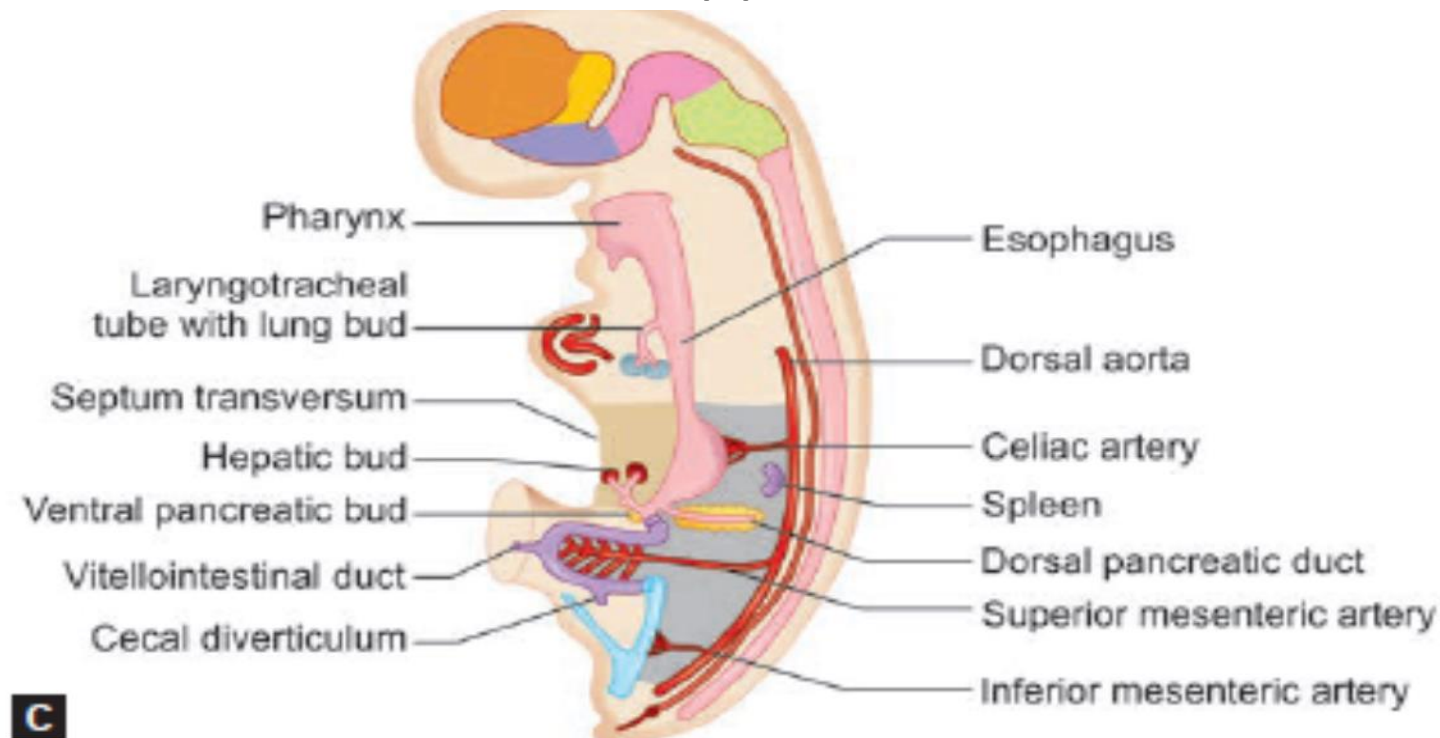
- foregut is in the head fold of the embryo
- separated from the stomodeum by the **buccopharyngeal membrane**
- hindgut is in the tail fold of the embryo
- hindgut is separated from the proctodeum by the ***cloacal membrane***
- buccopharyngeal & cloacal membranes disappear
- foregut & hindgut are in communication with stomodeum & proctodeum respectively
- gut communicates with the exterior



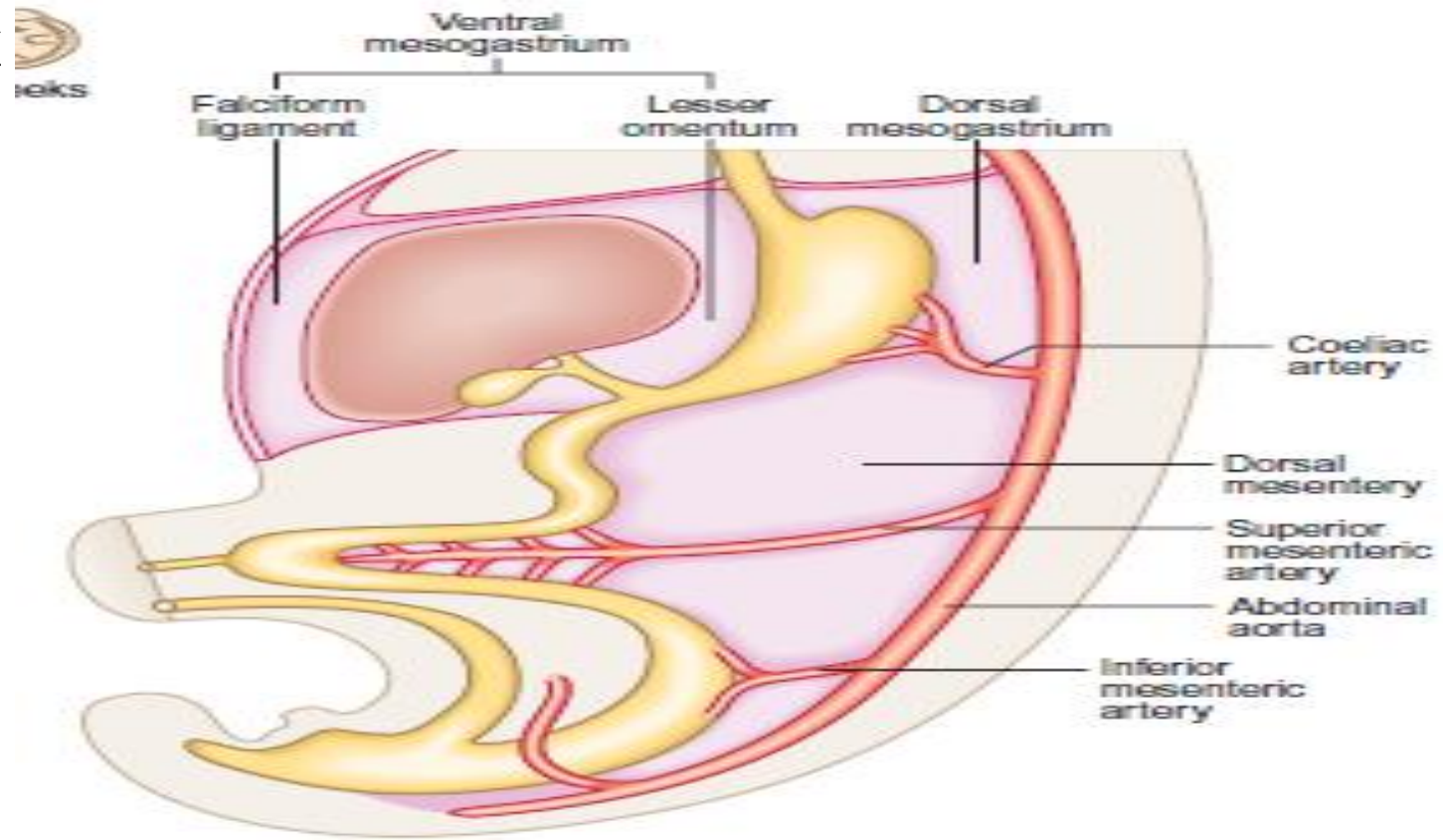
- midgut in early embryonic period communicates with extraembryonic part of yolk sac via ***vitellointestinal duct***
- vitellointestinal duct disappears by 5th week of development
- region of mouth & anal canal epithelium is derived from ectoderm of stomodeum & proctodeum respectively



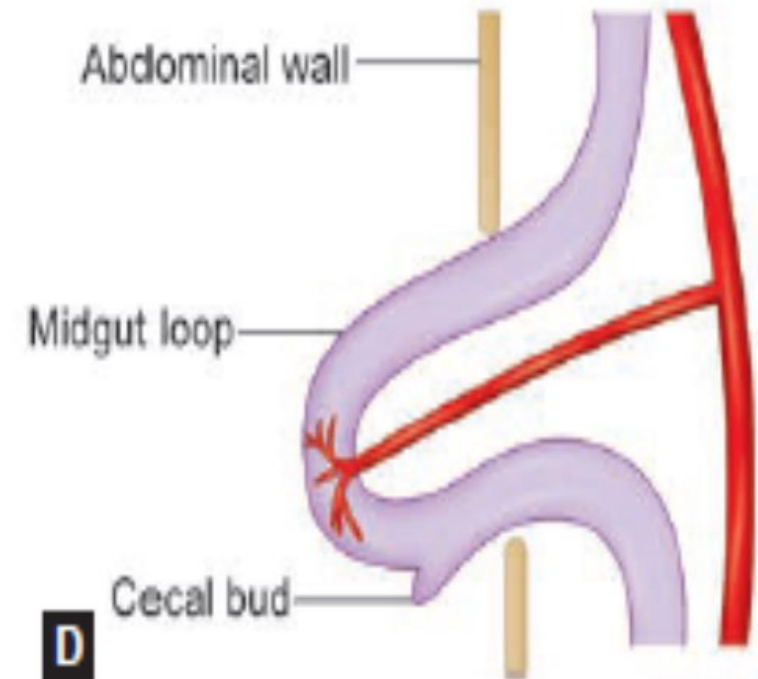
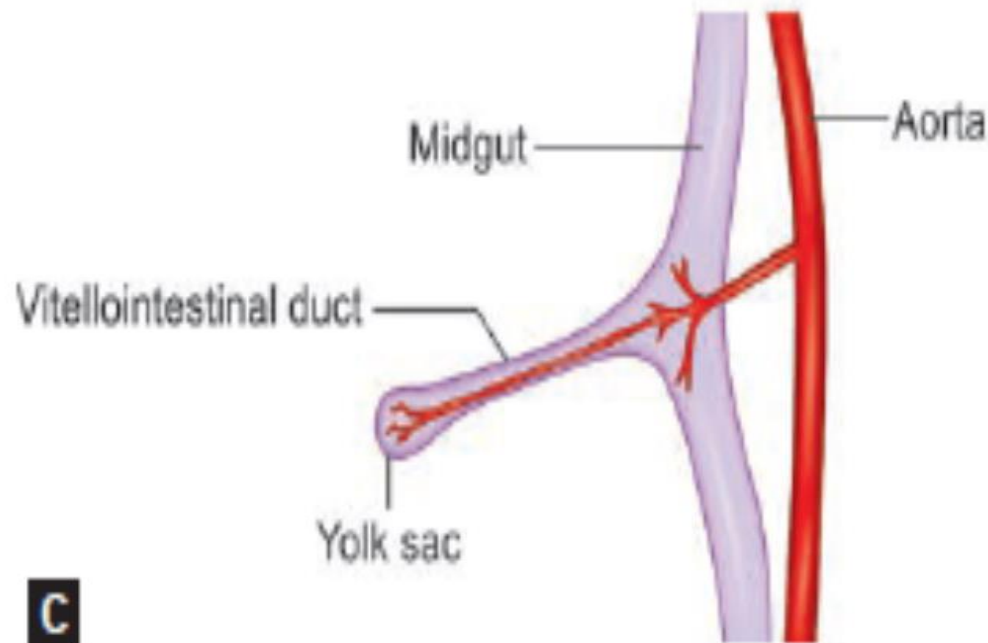
- gut is fixed to the ventral & dorsal body wall of the embryo by ***ventral & dorsal mesenteries***
- midline artery, ***dorsal aorta***, is established & comes to lie dorsal to the gut
- gives off a series of branches to the gut
- Those in the region of the midgut, run right up to the yolk sac are ***vitelline arteries***
- ventral branches of the dorsal aorta disappear & 3 remain



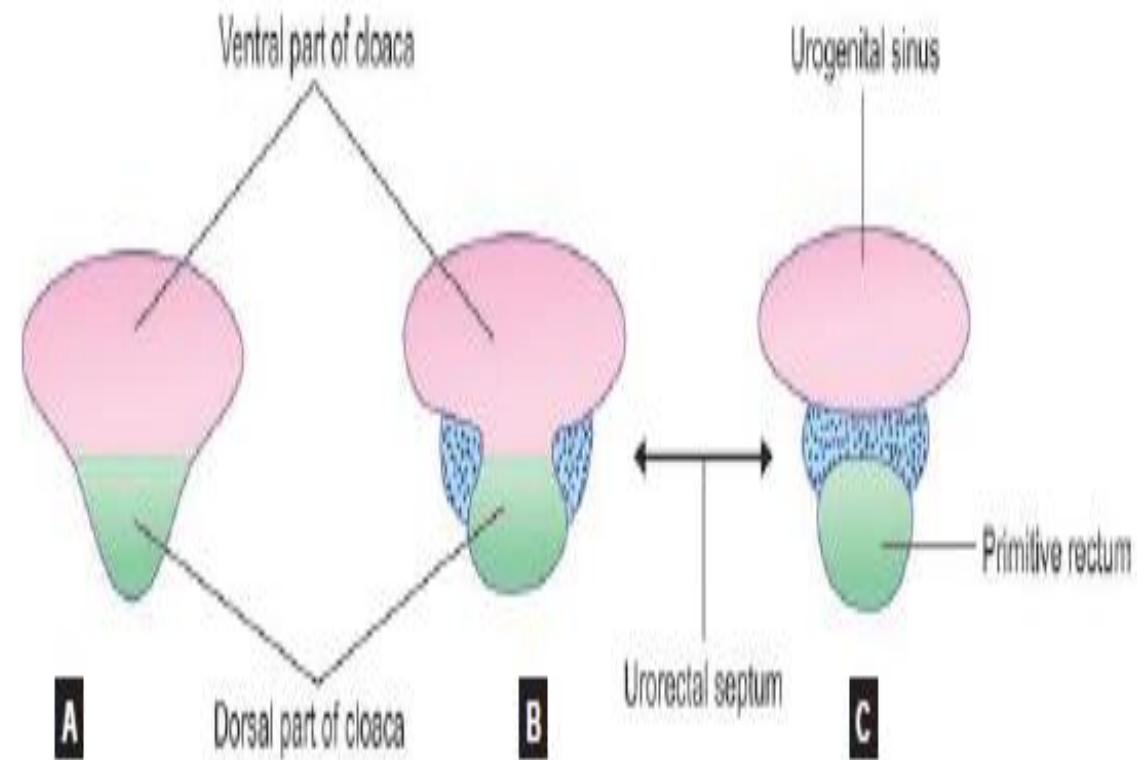
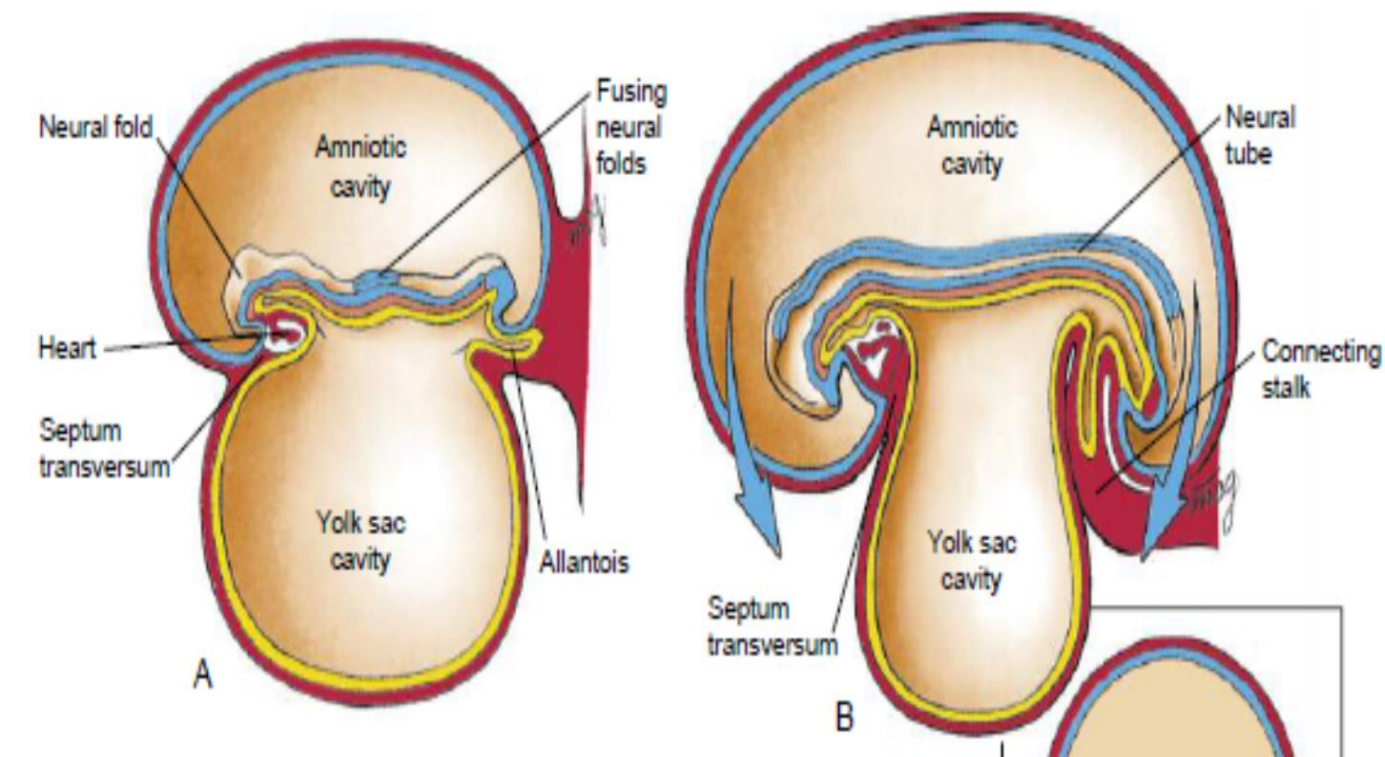
- artery of the abdominal part of the foregut – ***celiac artery***
- midgut - ***superior mesenteric artery***
- hindgut - ***inferior mesenteric artery***
- communication between the yolk sac & the midgut is narrowed down the midgut becomes tubular
- midgut assumes the form of a loop ,superior mesenteric artery runs in the mesentery of this loop to its apex



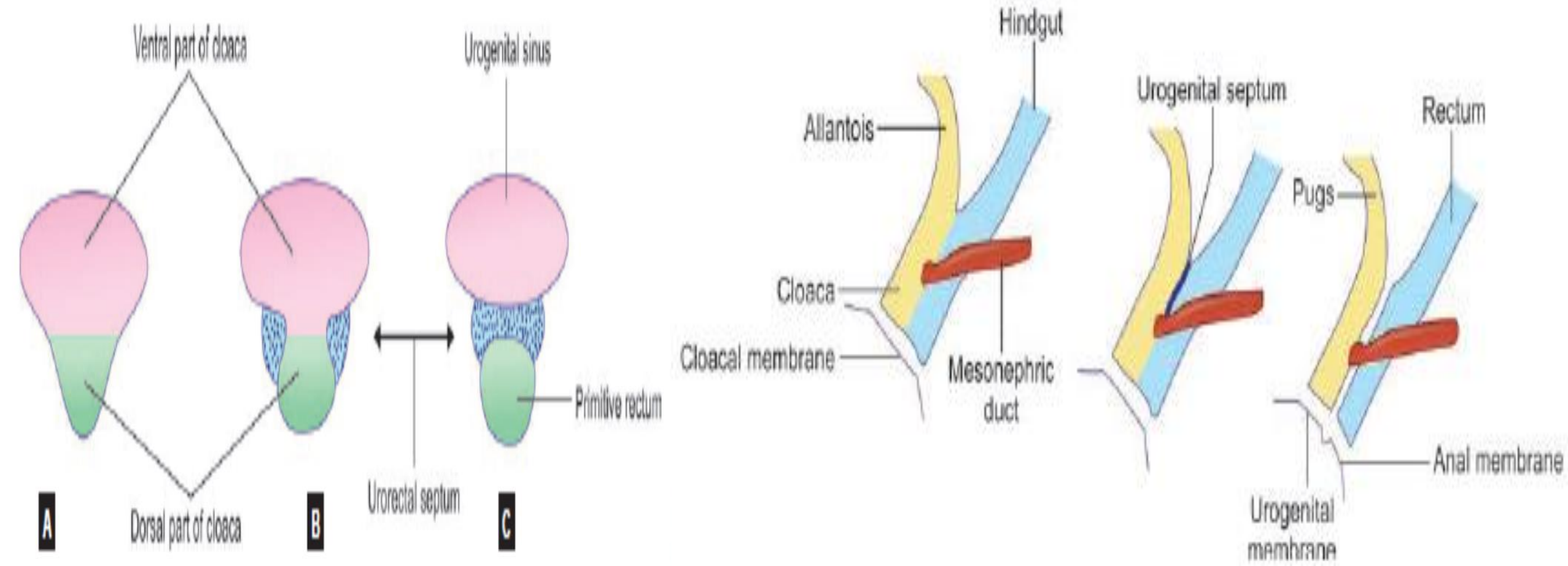
- loop has a *proximal* or *prearterial* segment & *distal* or *postarterial* segment
- *cecal bud* arises from the postarterial segment near the apex of the loop
- For a number of weeks, midgut loop comes to lie outside the abdominal cavity of the embryo
- Passes through the umbilical opening into a part of the extraembryonic coelom that persists in relation to the most proximal part of the umbilical cord
- loop is withdrawn into the abdominal cavity



- formation of Allantoic diverticulum, opens into the ventral aspect of the hindgut
- part of the hindgut caudal to the attachment of Allantoic diverticulum- ***cloaca***
- Cloaca divide into broad ventral part & narrow dorsal part
- formed by ***urorectal septum***, formed in the angle between the allantois & cloaca
- ventral subdivision - ***primitive urogenital sinus***
- gives origin to some parts of the urogenital system



- dorsal part -***primitive rectum*** forms rectum & part of anal canal
- urorectal septum grows toward the cloacal membrane and fuses with it
- cloacal membrane is divided into a :-
- ventral *urogenital membrane*, related to the urogenital sinus
- dorsal *anal membrane* related to the rectum



- Mesoderm around the anal membrane becomes heaped up results in the anal membrane lie at the bottom of anal pit, or proctodeum
- anal pit contributes to the formation of the anal canal

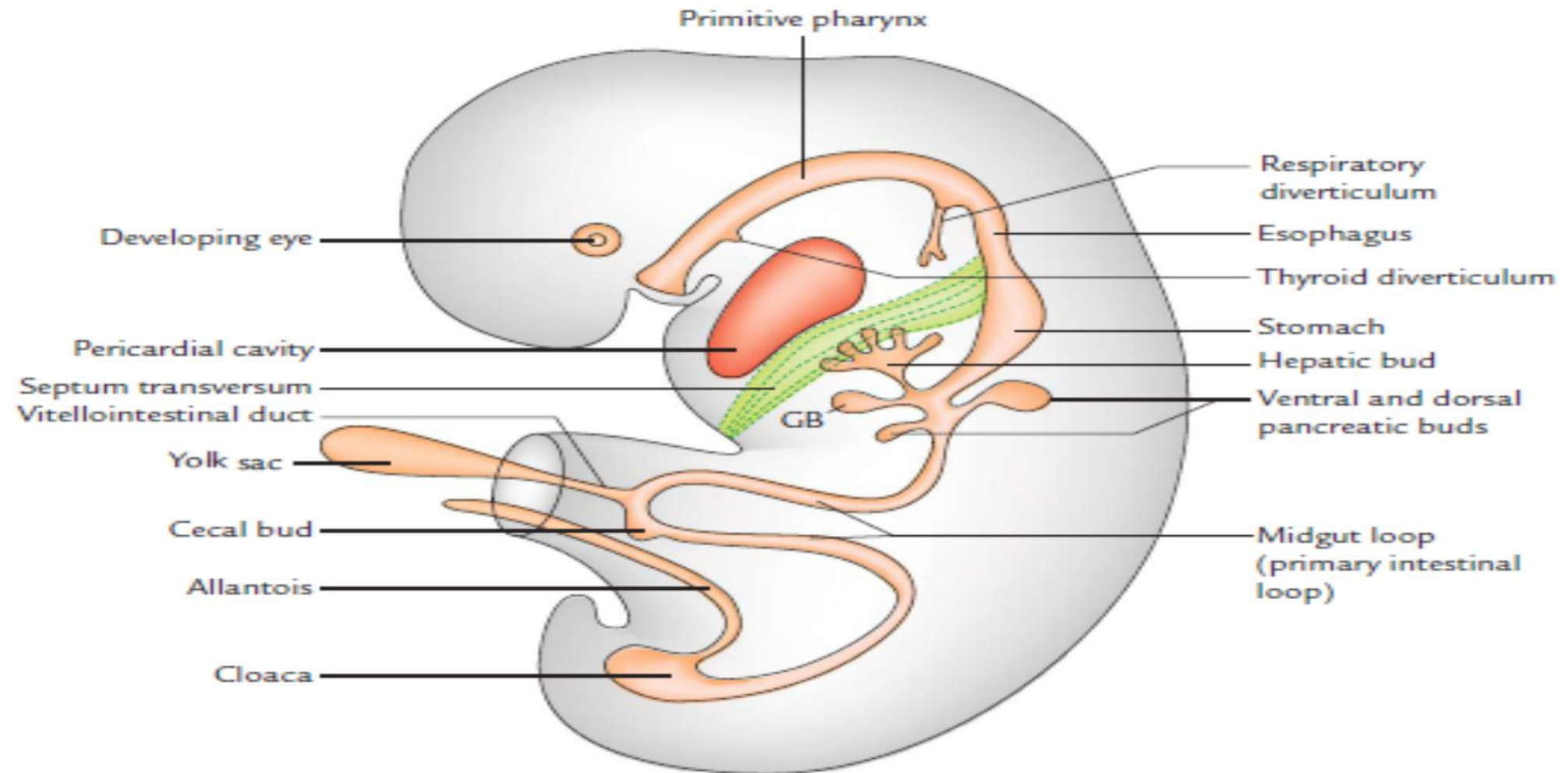


Fig. 13.4 Schematic diagram of 5-mm embryo showing the formation of the digestive tract. Note the subdivisions of digestive tract into foregut, midgut, and hindgut, and various derivatives originating from their endoderm. GB = gallbladder.

- The derivatives of the foregut are :-
 - A. primordial pharynx & its derivatives
 - B. lower respiratory system
 - C. esophagus and stomach
 - D. duodenum, distal to the opening of the bile duct
 - E. liver, biliary apparatus (hepatic ducts, gallbladder, and bile duct), and pancreas

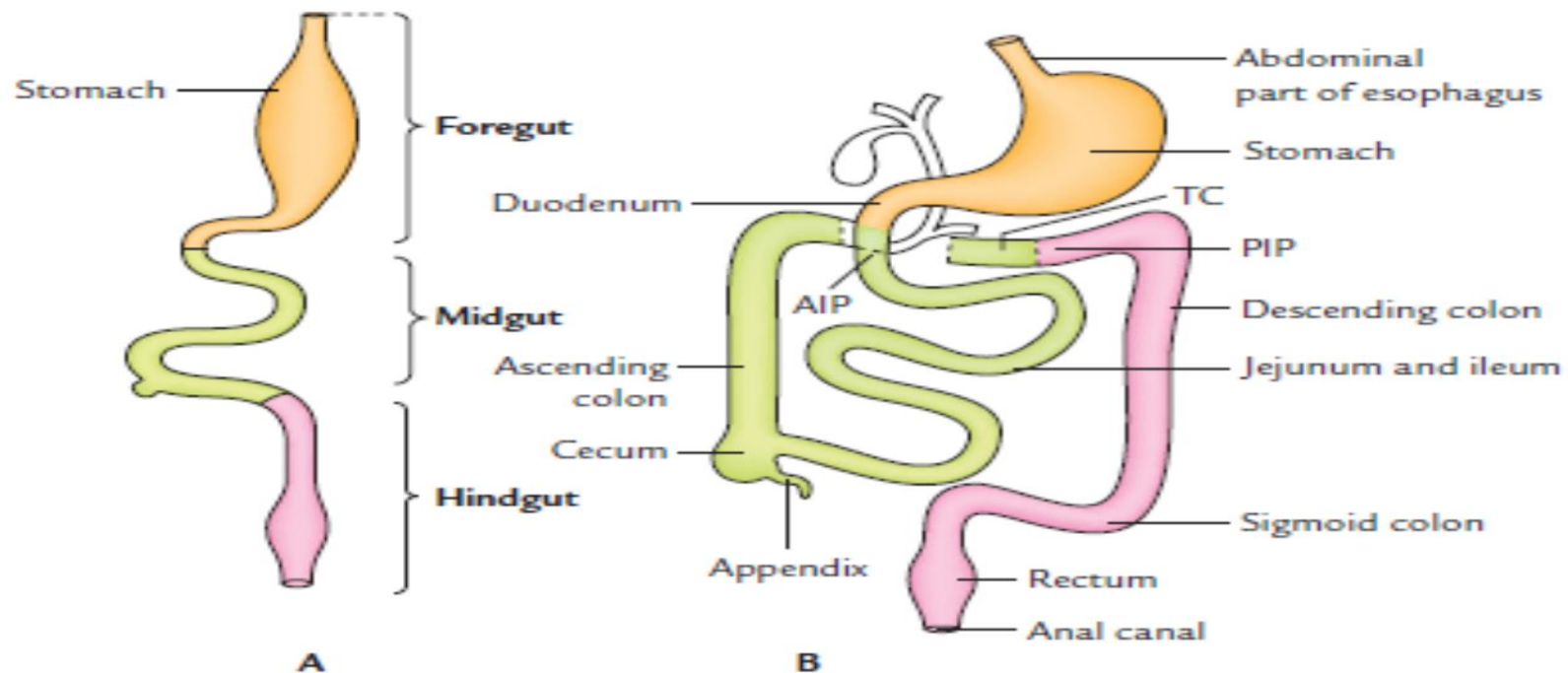


Fig. 13.5 Derivatives of various abdominal parts of the gut. A. Primitive gut. B. Adult gut. TC = transverse colon; AIP = anterior intestinal portal; PIP = posterior intestinal portal.

- **Foregut Development**

- *Esophagus*

- develops from foregut caudal to pharynx
- is at first short
- elongates with formation of the neck, descent of the diaphragm & enlargement of the pleural cavities
- reaches final relative length by the 7th week
- epithelium and glands are derived from endoderm
- epithelium proliferates and obliterates the lumen
- recanalization of the esophagus normally occurs by the end of the 8th week

- musculature of the esophagus is derived from splanchnic mesenchyme surrounding the foregut
- upper two-thirds of the esophagus, the mesenchyme forms striated muscle
- Middle one-third forms mixed striated and smooth
- lower one-third, the muscle formed is smooth

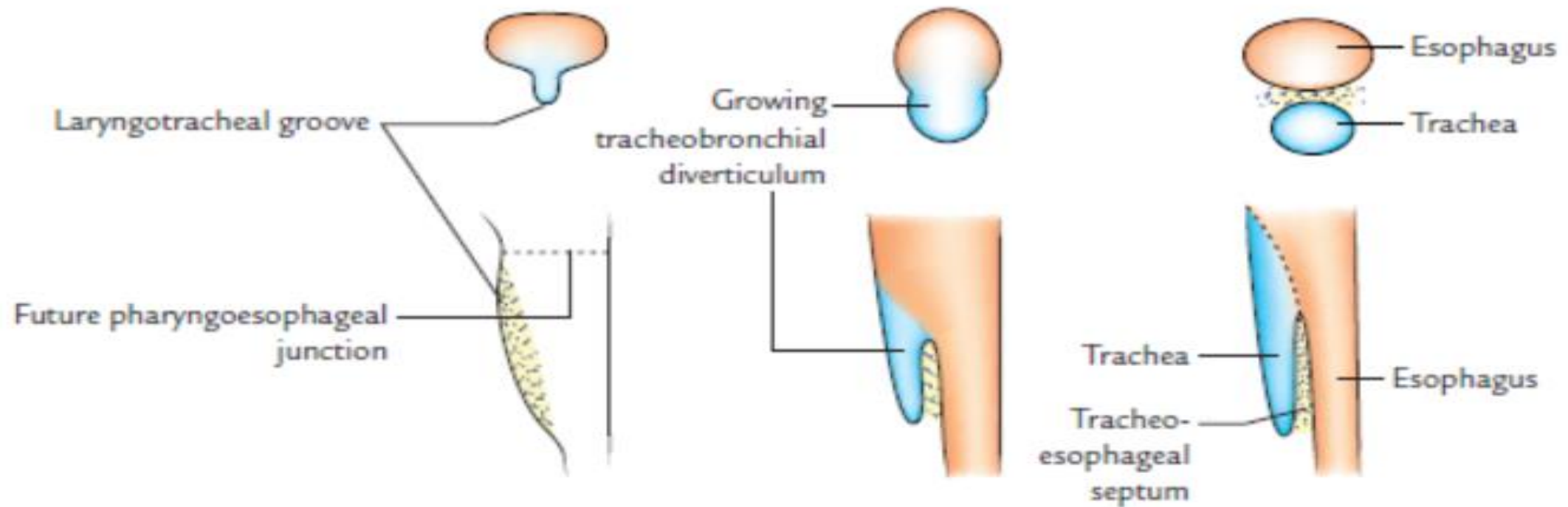
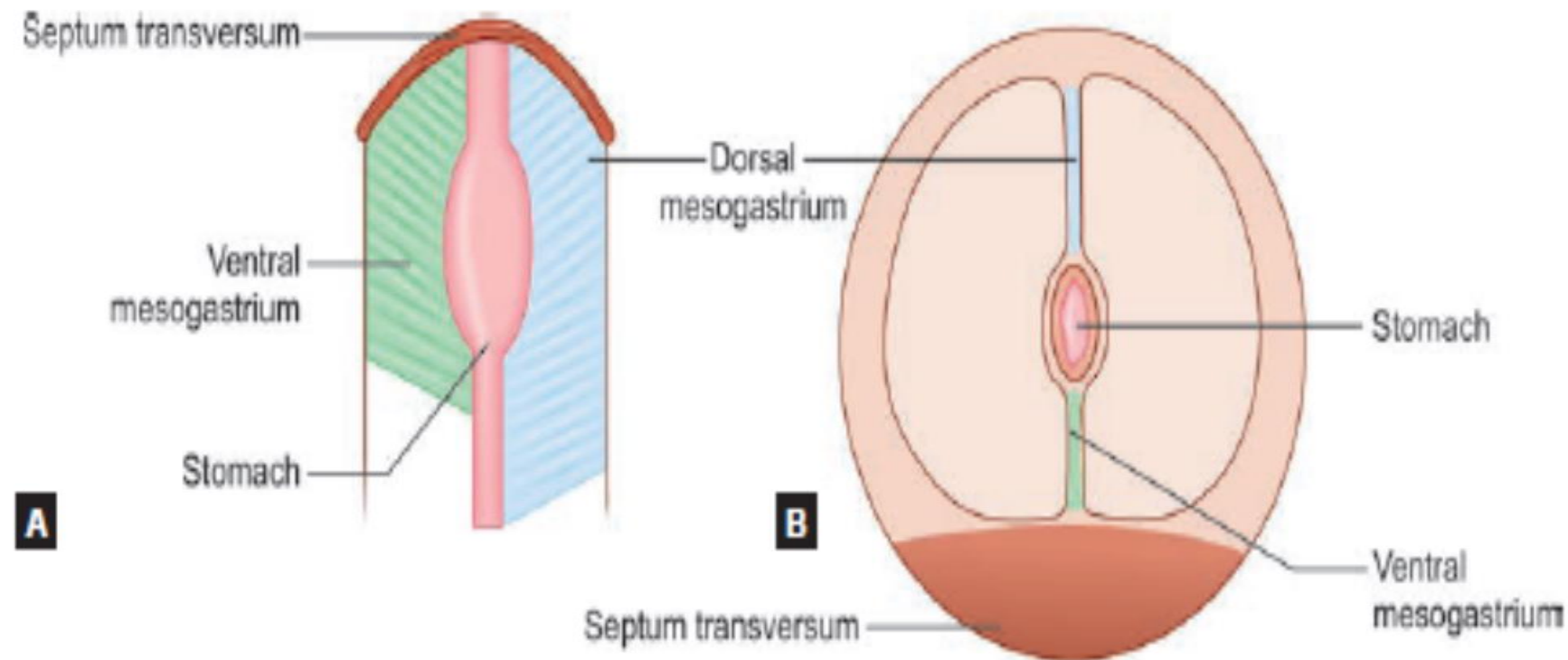


Fig. 13.6 Development of esophagus.

- ***Stomach***

- first seen as a fusiform dilatation of foregut distal to esophagus in the 4th week
- dorsal border is attached to posterior abdominal wall by ***dorsal mesogastrium***
- ventral border is attached to septum transversum - ***ventral mesogastrium***
- Liver & diaphragm are formed in the substance of the septum transversum
- ventral mesogastrium passes from stomach to liver & from liver to diaphragm & anterior abdominal wall



Septum transversum

Ventral
mesogastrium

Stomach

Dorsal
mesogastrium

Stomach

Ventral
mesogastrium

Septum transversum

Septum transversum

Hindgut

Pharynx

Thoracic esophagus

Respiratory diverticulum

Stomach

Liver cords

Dorsal pancreatic rudiment

Gallbladder

Ventral pancreatic rudiment

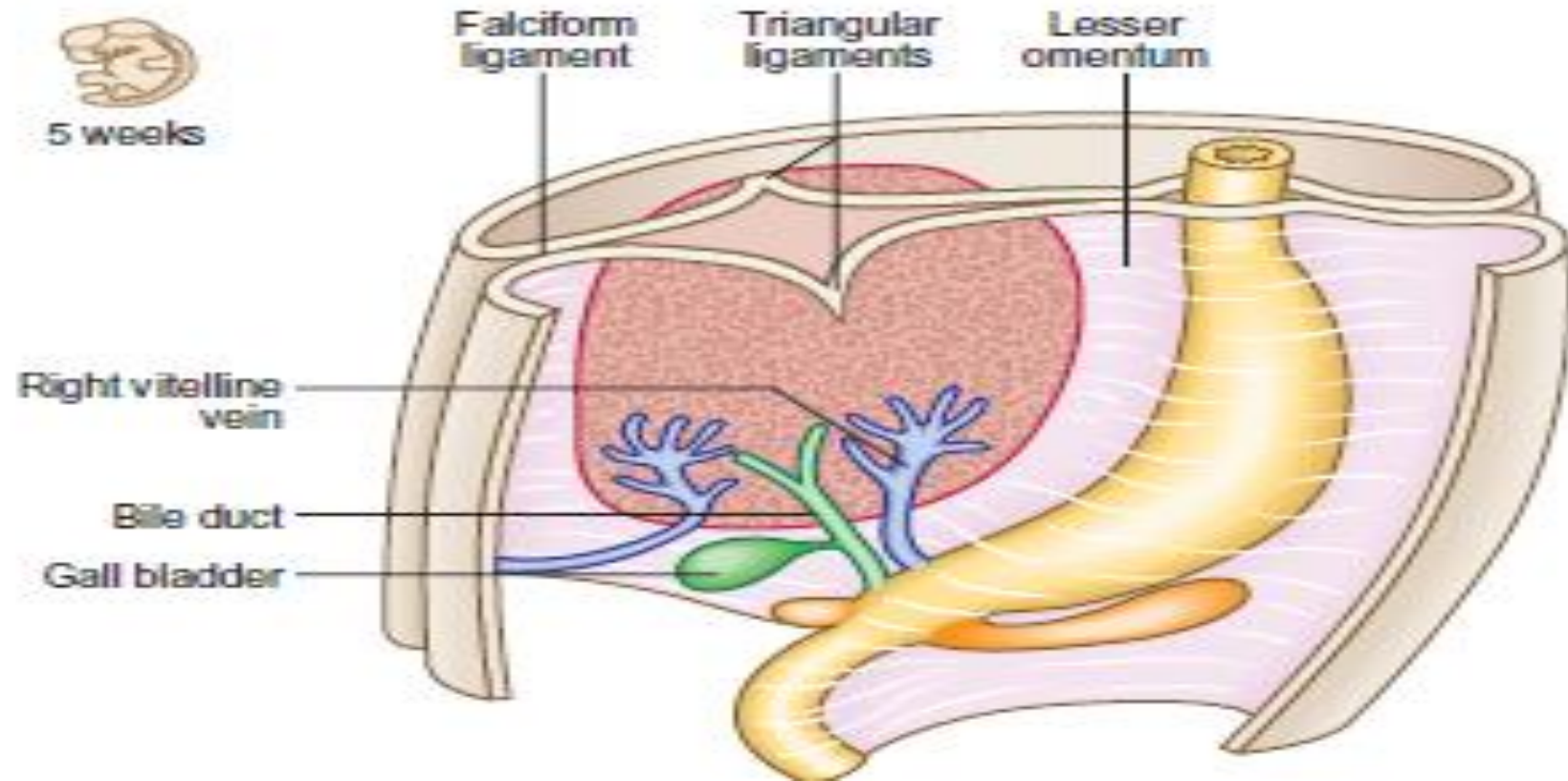
Ventral mesentery

Midgut

Dorsal mesentery

30 days

- ventral mesogastrium between liver & stomach becomes ***lesser omentum***
- Part between liver & diaphragm (and anterior abdominal wall) gives rise to ***coronary*** and ***falciform ligaments***
- dorsal mesogastrium is divided by the development of the spleen:-
- part between stomach & spleen is ***gastrosplenic ligament***
- part between spleen and posterior abdominal wall is ***lienorenal ligament***



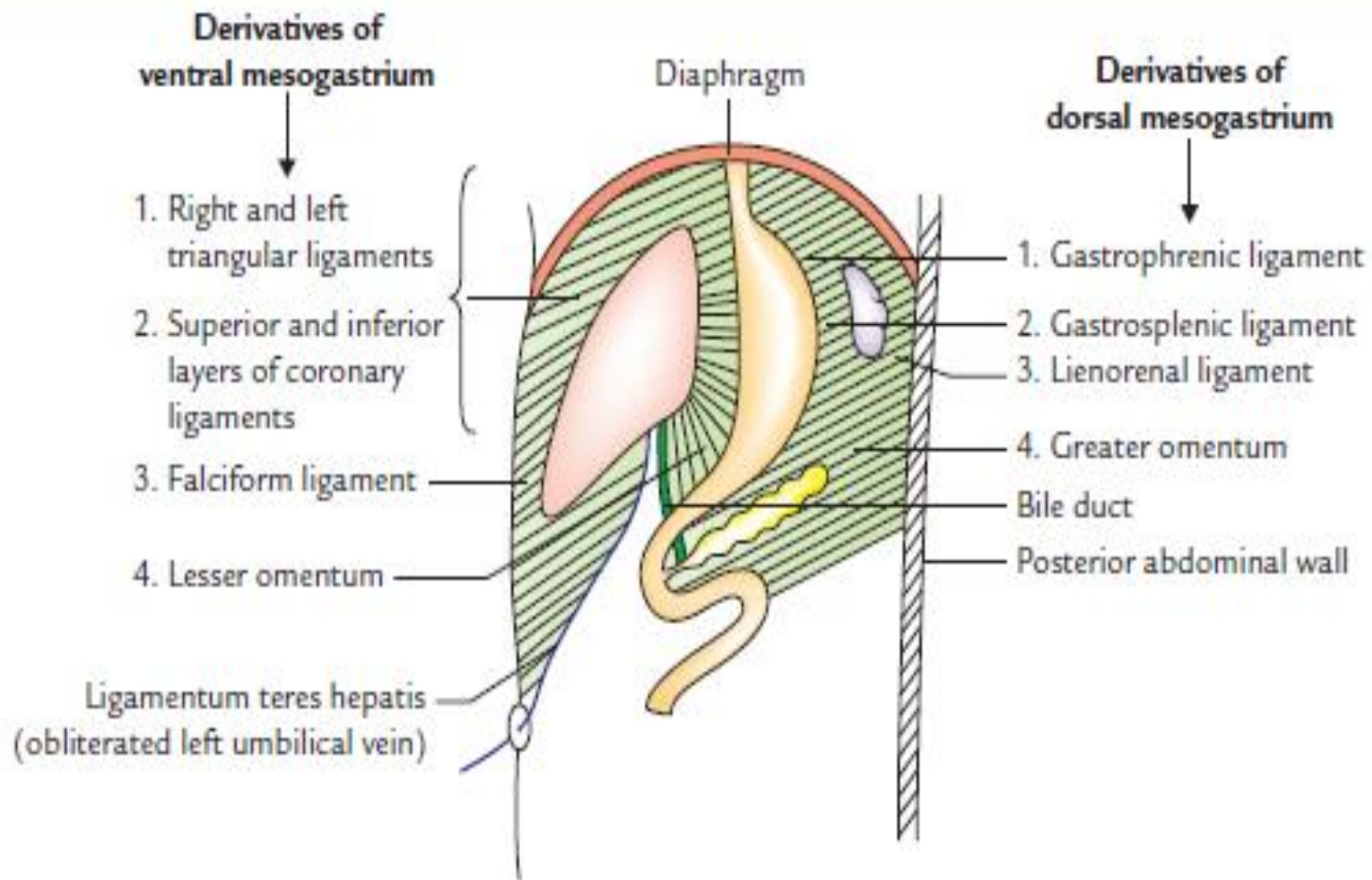
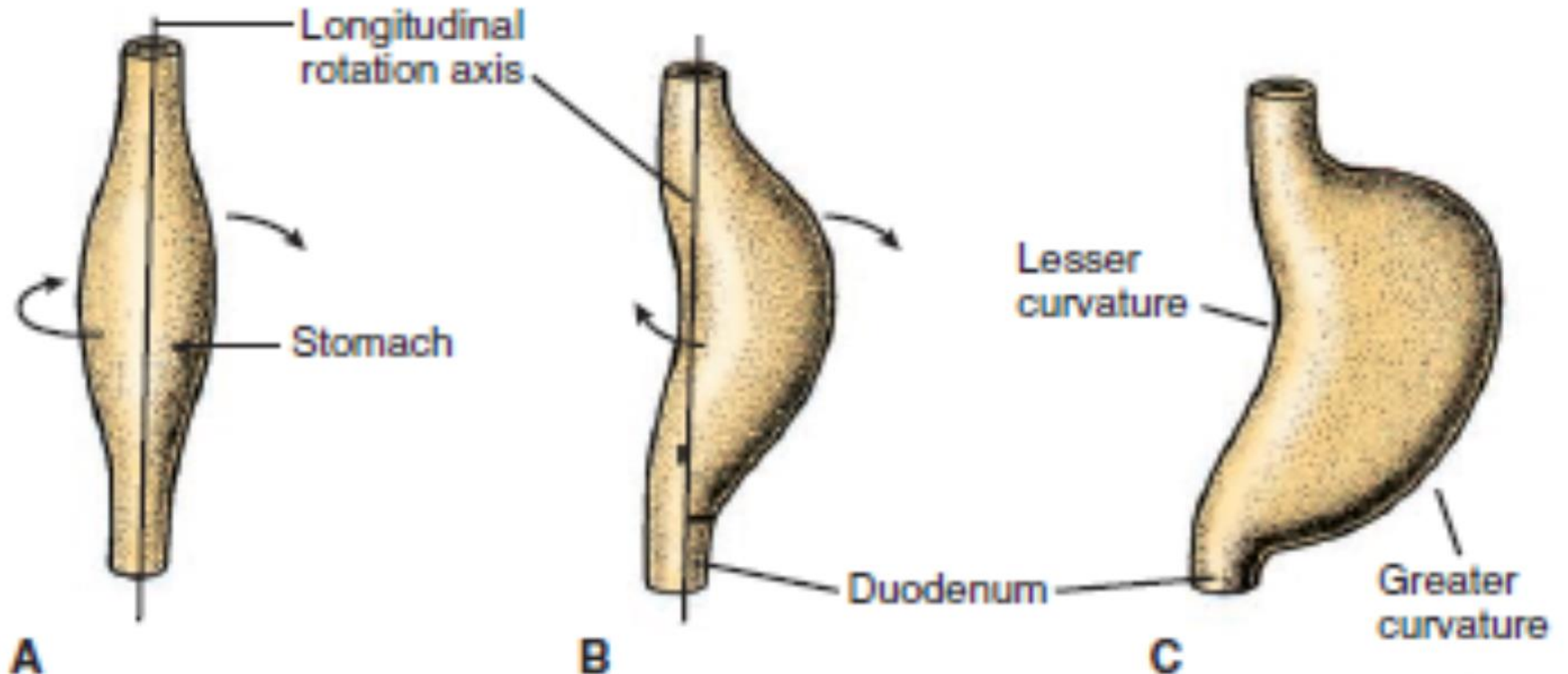
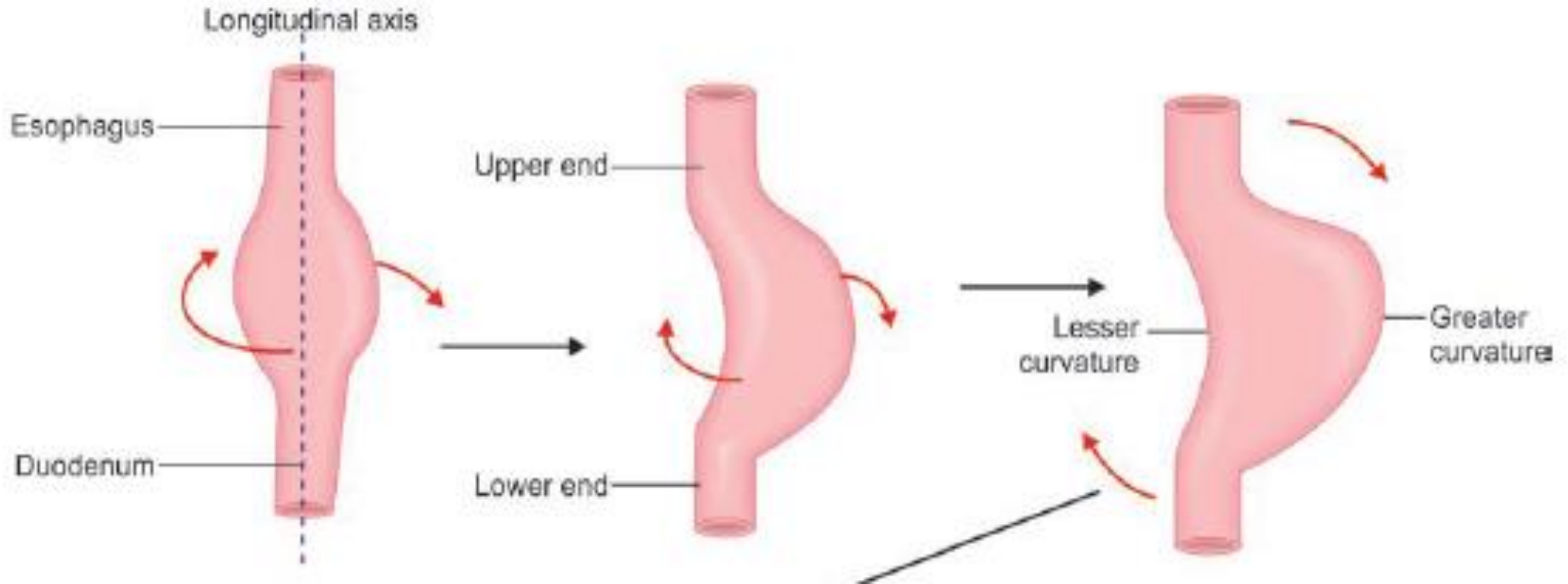


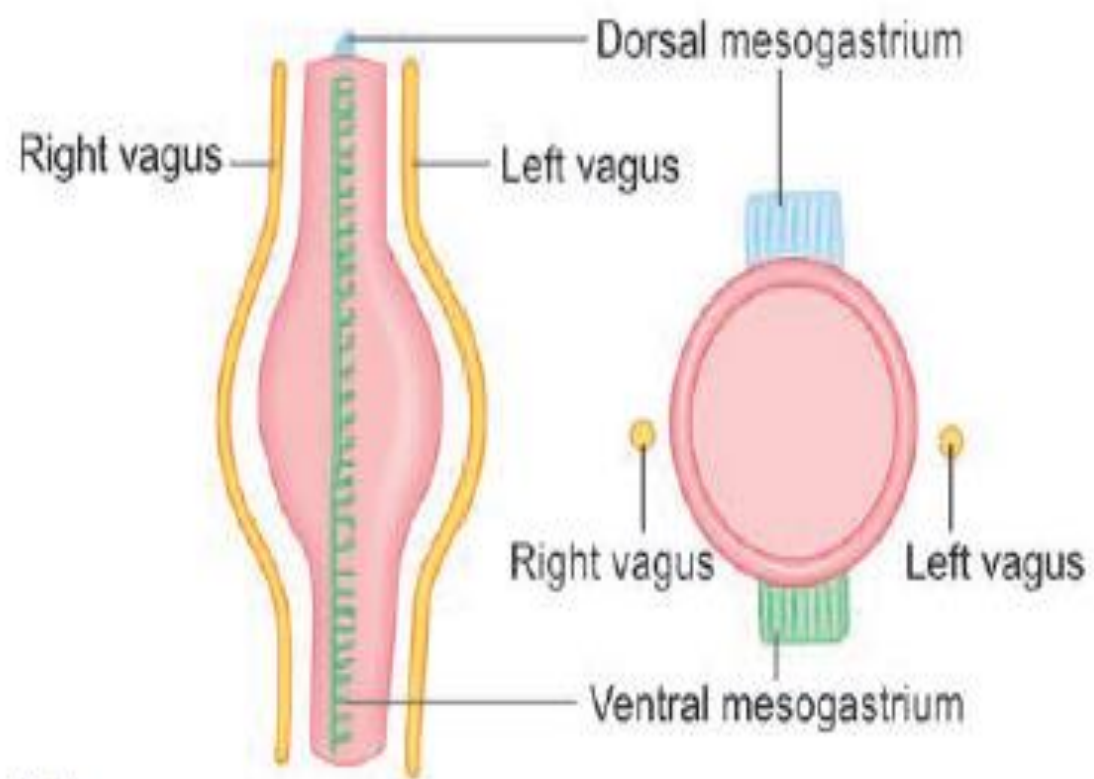
Fig. 13.10 Derivatives of ventral and dorsal mesogastria. Layers of coronary, and right and left triangular ligaments are not shown.

- rotation of the stomach can be explained as follows:-
- First rotation is 90° clockwise along longitudinal axis
- changes orientation of its surfaces and the position of vagus nerves
- Second rotation: in transverse/anteroposterior axis
- changes position fundus & duodenum and in the position of ends of stomach

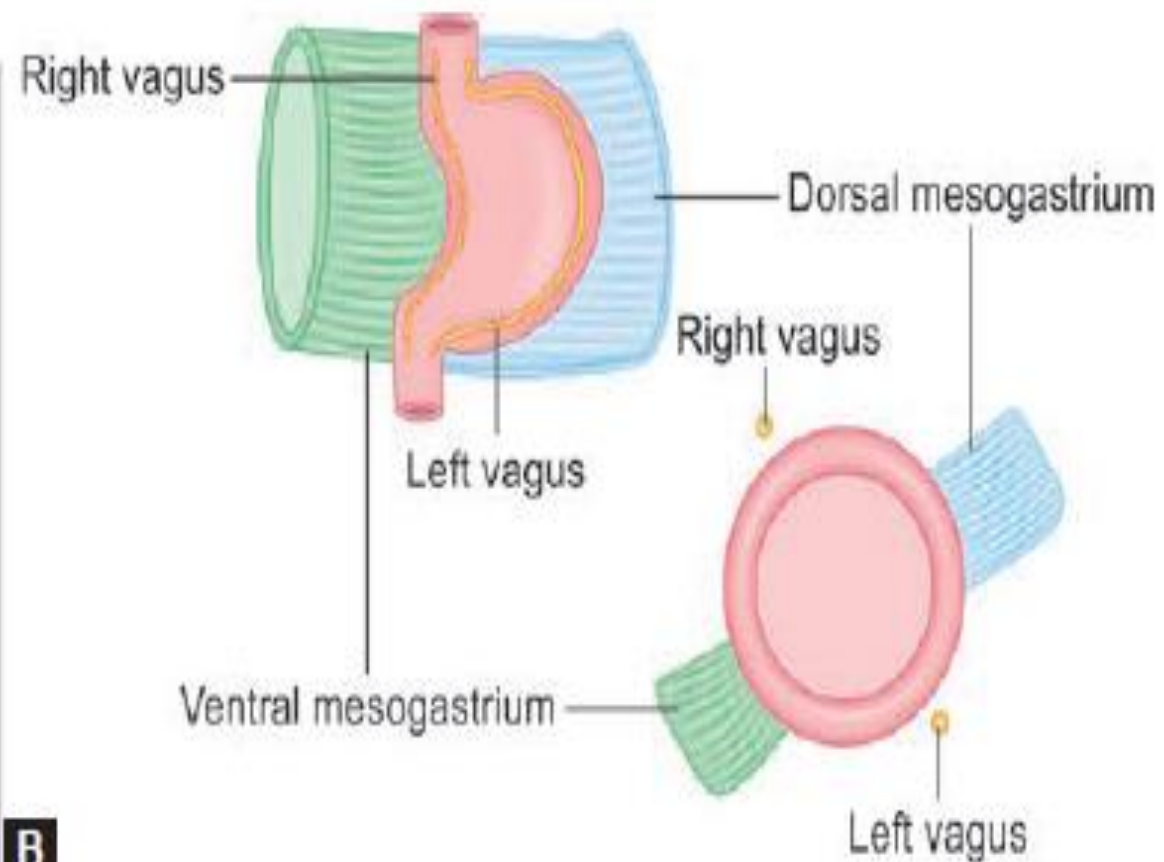


- it undergoes differential growth results in *alteration in its shape & orientation*
- original ventral border comes to face upward & to the right & becomes ***lesser curvature***
- dorsal border points downward & to the left and becomes ***greater curvature***
- original left surface becomes anterior surface
- original right surface becomes posterior surface





A



B

- rotation and differential growth of surfaces change relationship of right and left vagus nerves to posteroinferior and anterosuperior surfaces respectively
- During rotation cranial end tilts to the left & caudal end to the right to assume the adult position
- Rotation and disproportionate growth of the stomach alters the position of dorsal and ventral mesogastria

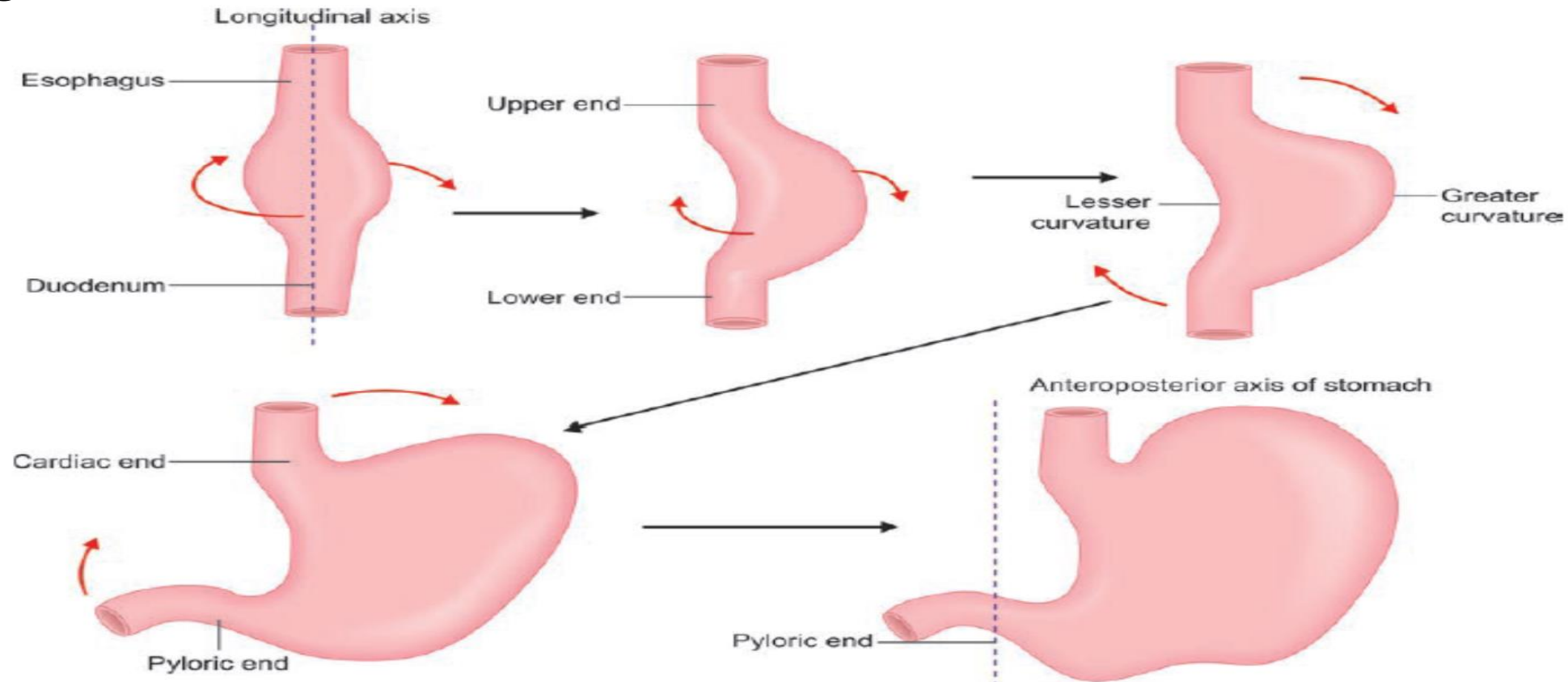
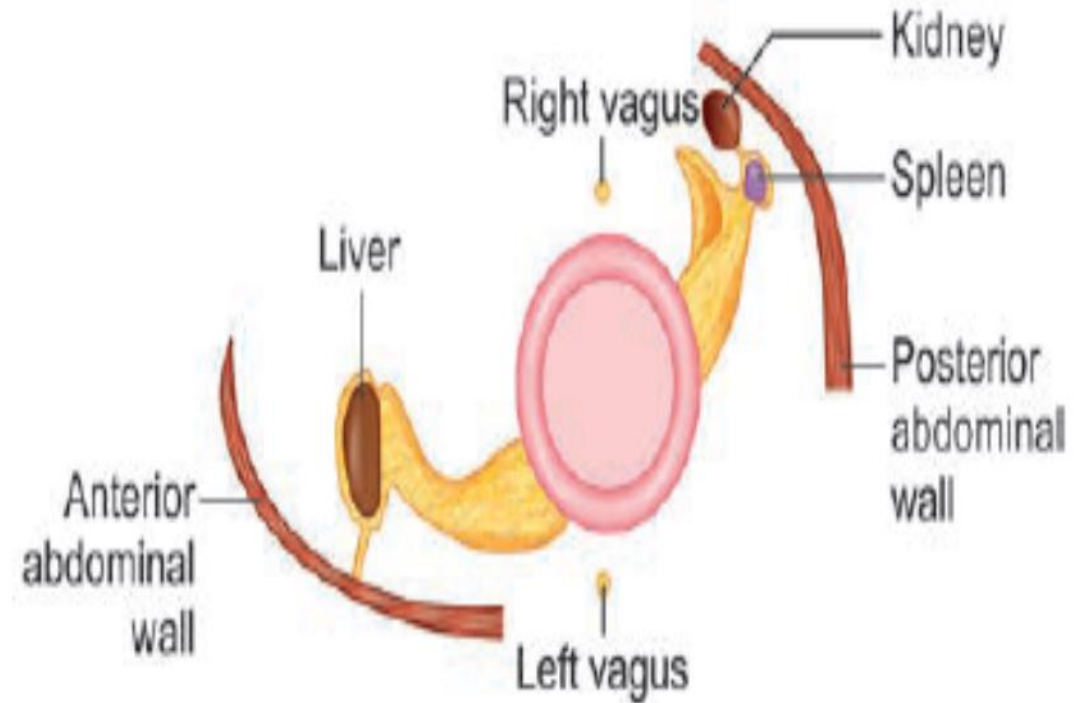
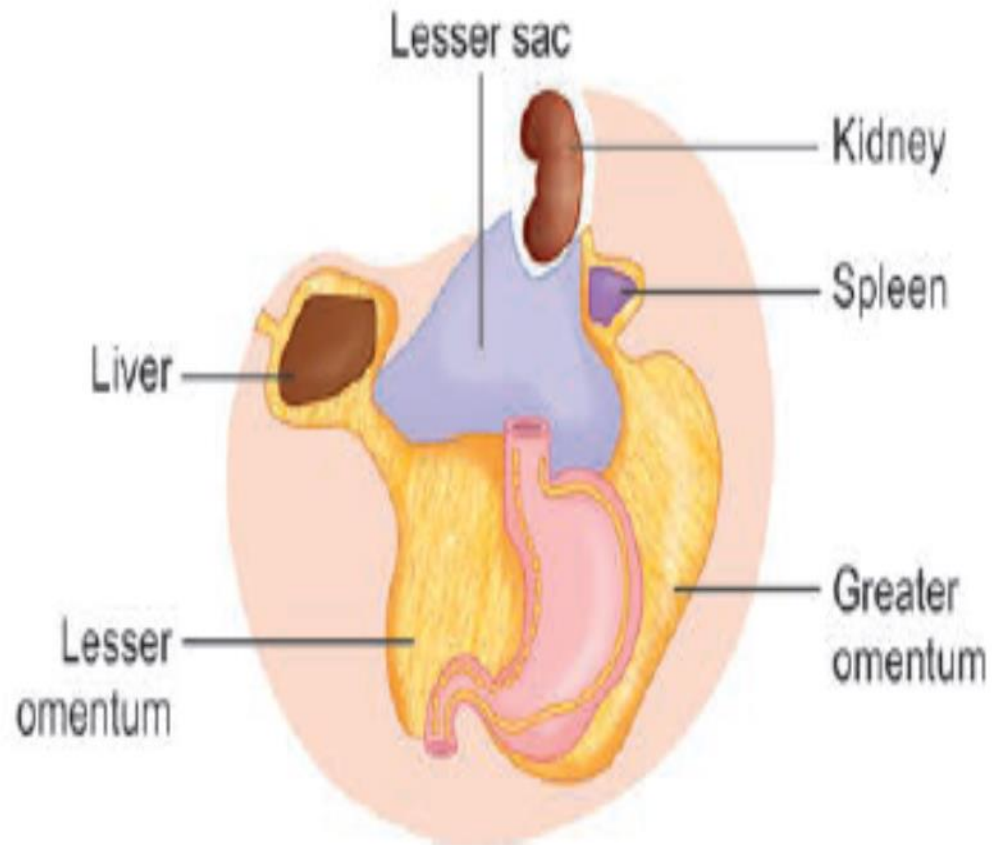


Fig. 13.8: Changes in the position and shape of stomach

- Rotation along longitudinal axis pulls dorsal mesogastrium to left forms the *lesser sac/omental bursa* behind the stomach
- the ventral mesogastrium is pulled to the right
- With the growth of stomach, the dorsal mesogastrium lengthens

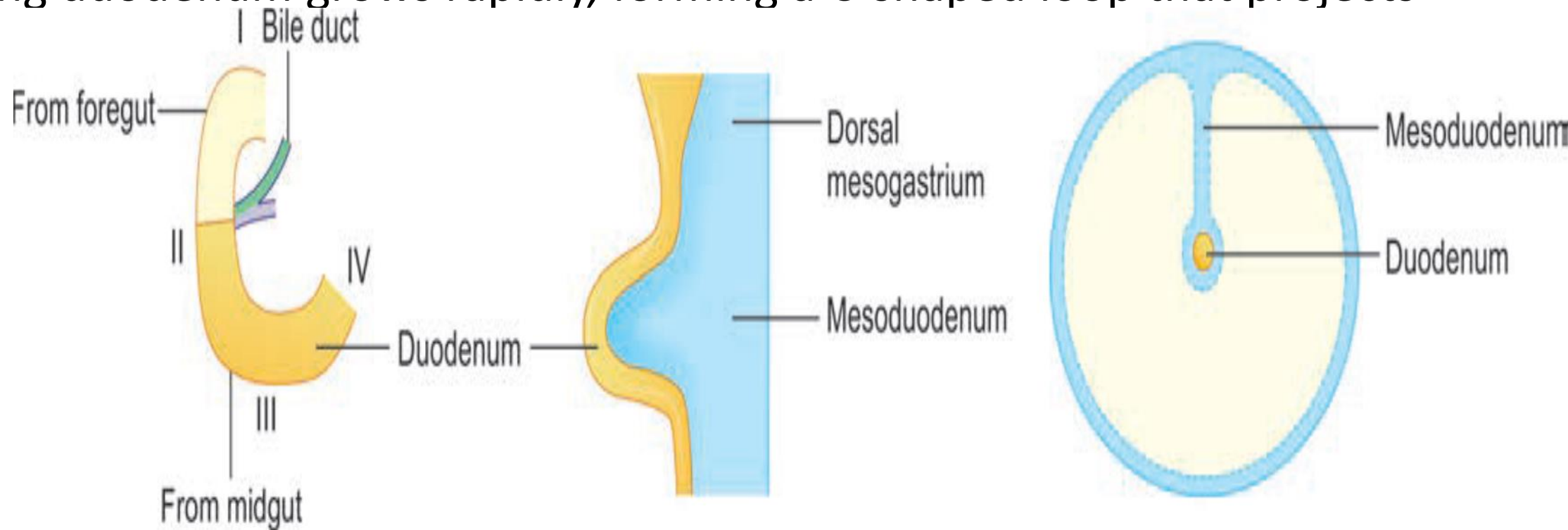


Figs 13.9A to C: Changes in the relation of vagus nerve and the mesogastrium to the developing stomach

- rotation along anteroposterior axis, dorsal mesogastrium bulges downward and continues to grow to form a double-layered greater omentum
- lesser omentum and falciform ligament are formed from the ventral mesogastrium (a derivative of septum transversum)

• **Development of the Duodenum**

- begins to develop 4th week
- superior (or first) part & upper half of the descending (or second) part derived from the foregut
- rest develops from most proximal part of the midgut
- junction of the two parts is just distal to the origin of the bile duct
- developing duodenum grows rapidly, forming a C-shaped loop that projects ventrally



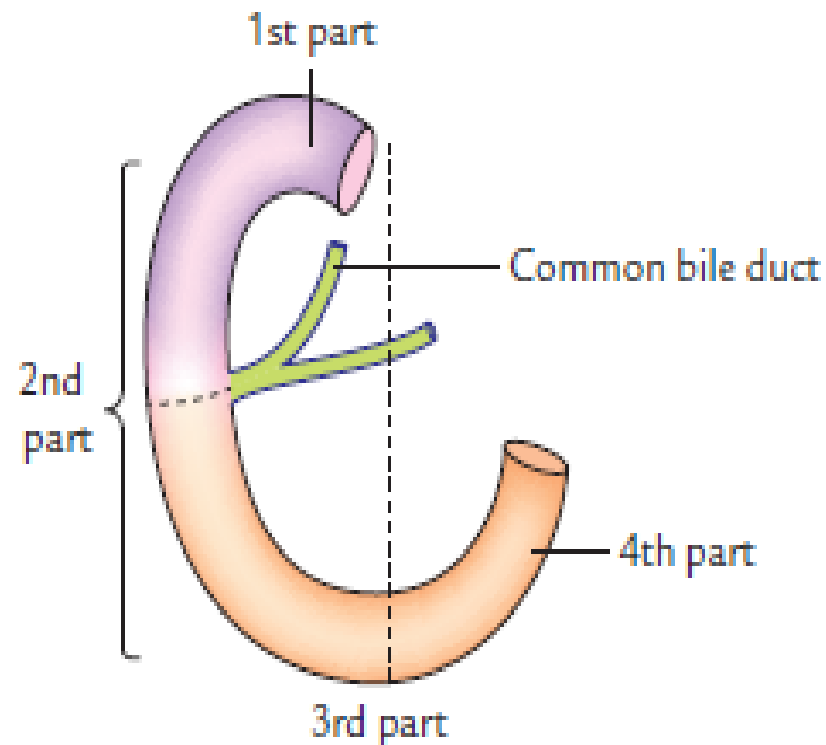


Fig. 13.12 Development of duodenum. Note, first part and second part up to the opening of common bile duct is derived from foregut (violet color). The second part of the duodenum (distal to opening of common bile duct) along with third and fourth parts is derived from midgut.

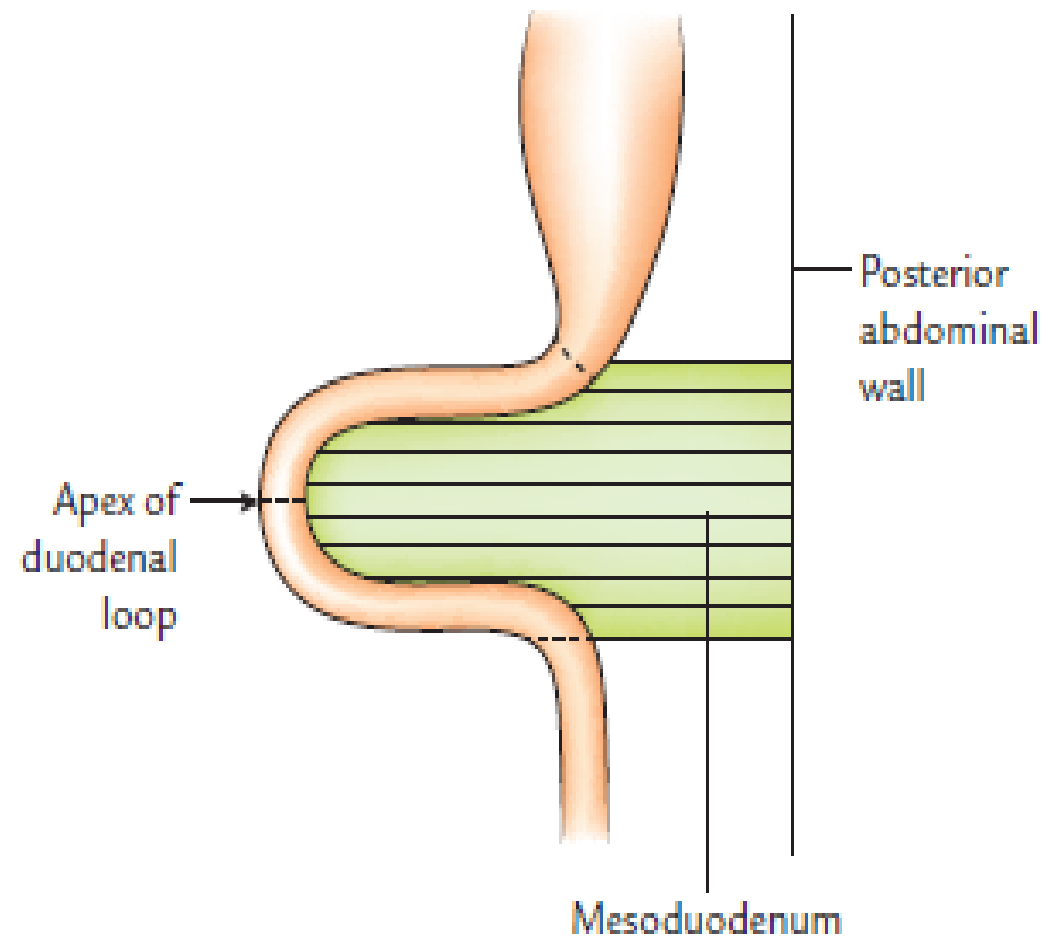
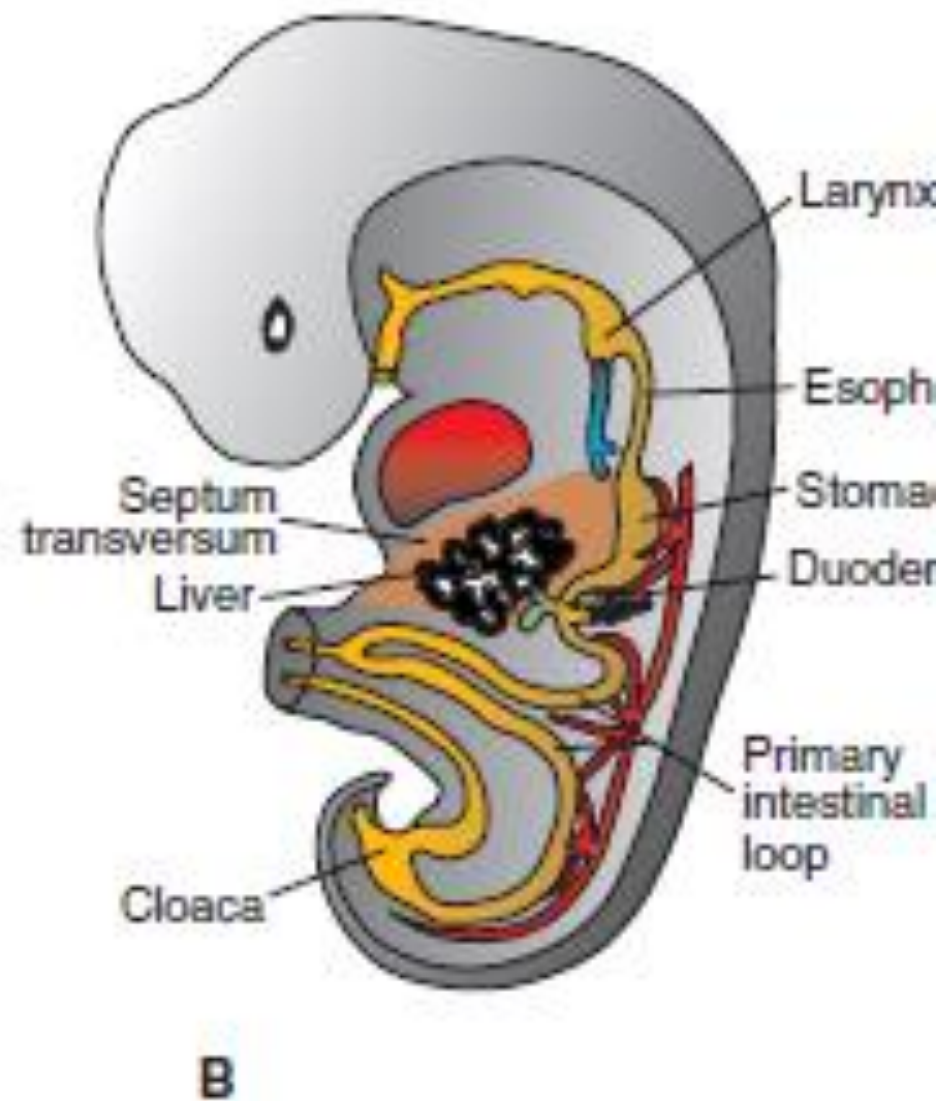
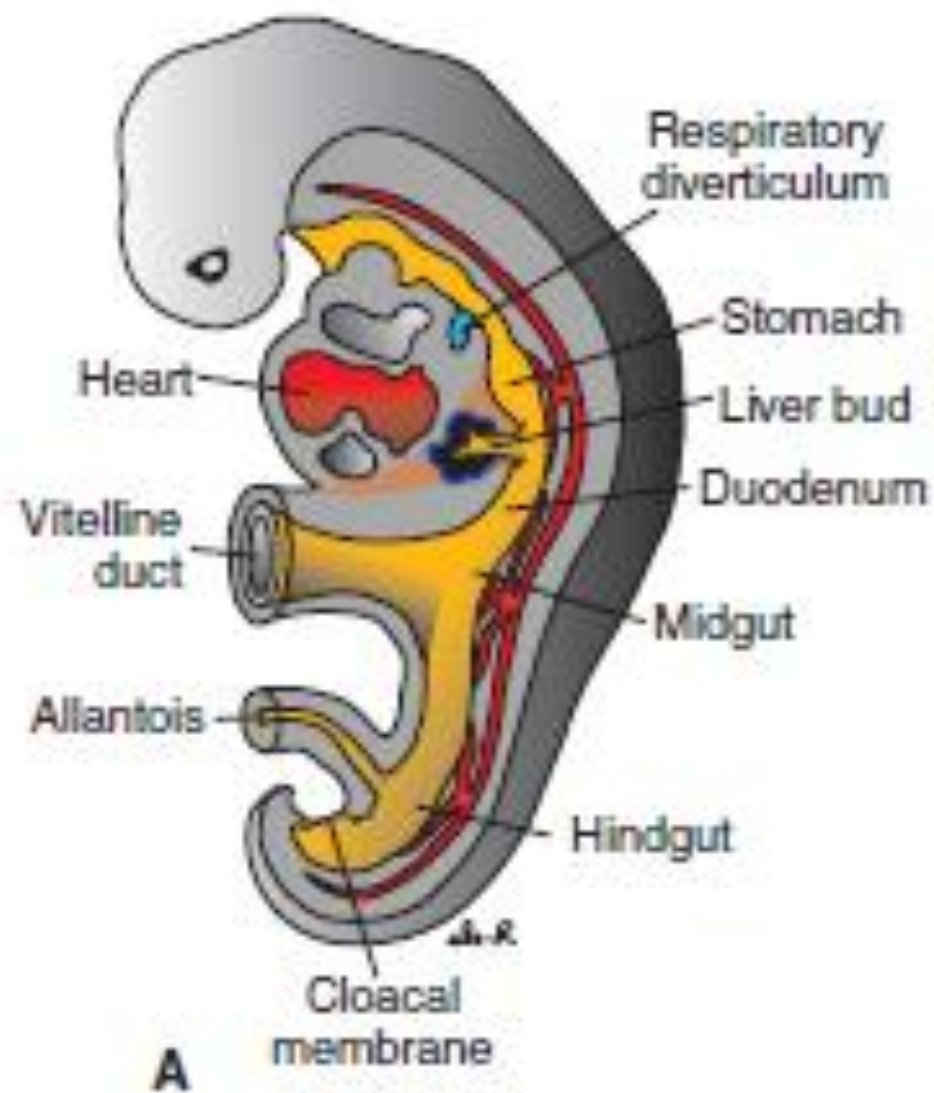


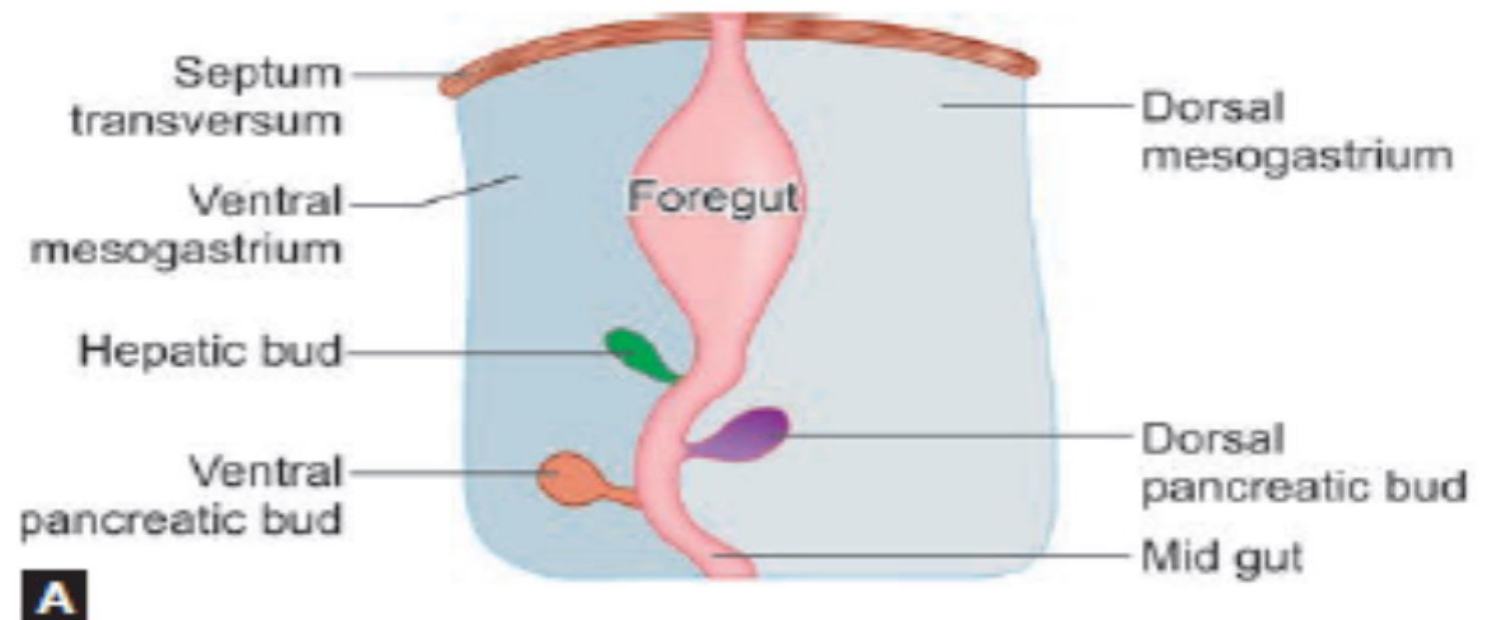
Fig. 13.13 Duodenal loop formed from parts of foregut and midgut. Note the mesoduodenum extending between duodenal loop and posterior abdominal wall.

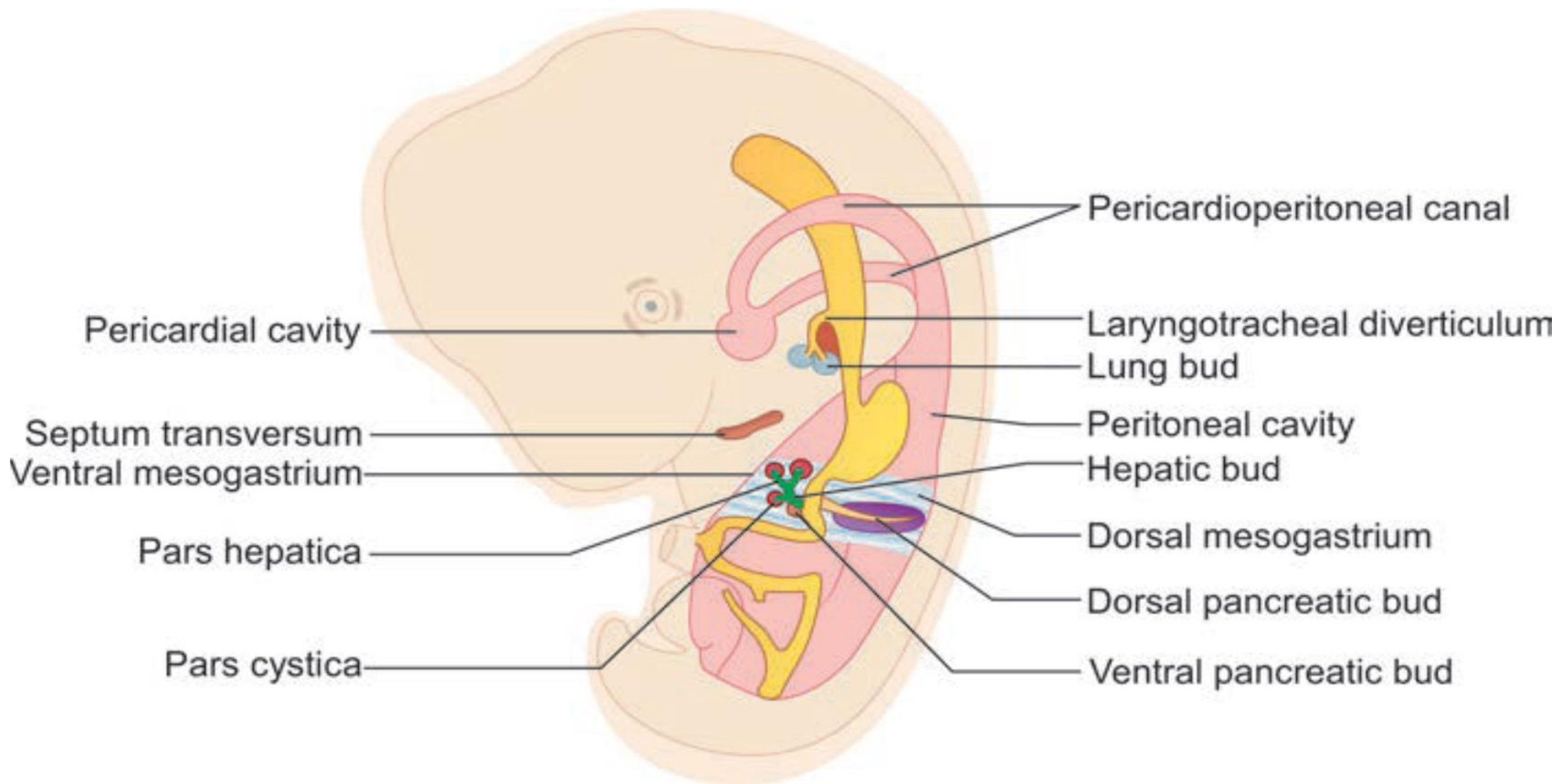
- part of gut that gives rise to the duodenum forms a loop attached to the posterior abdominal wall by a mesentery- ***mesoduodenum***
- As stomach rotates, duodenal loop rotates to the right
- mesoduodenum fuses with peritoneum of the posterior abdominal wall result most of duodenum becomes retroperitoneal
- mesoduodenum persists as small part of the duodenum adjacent to the pylorus
- part seen in radiographs as the duodenal cap



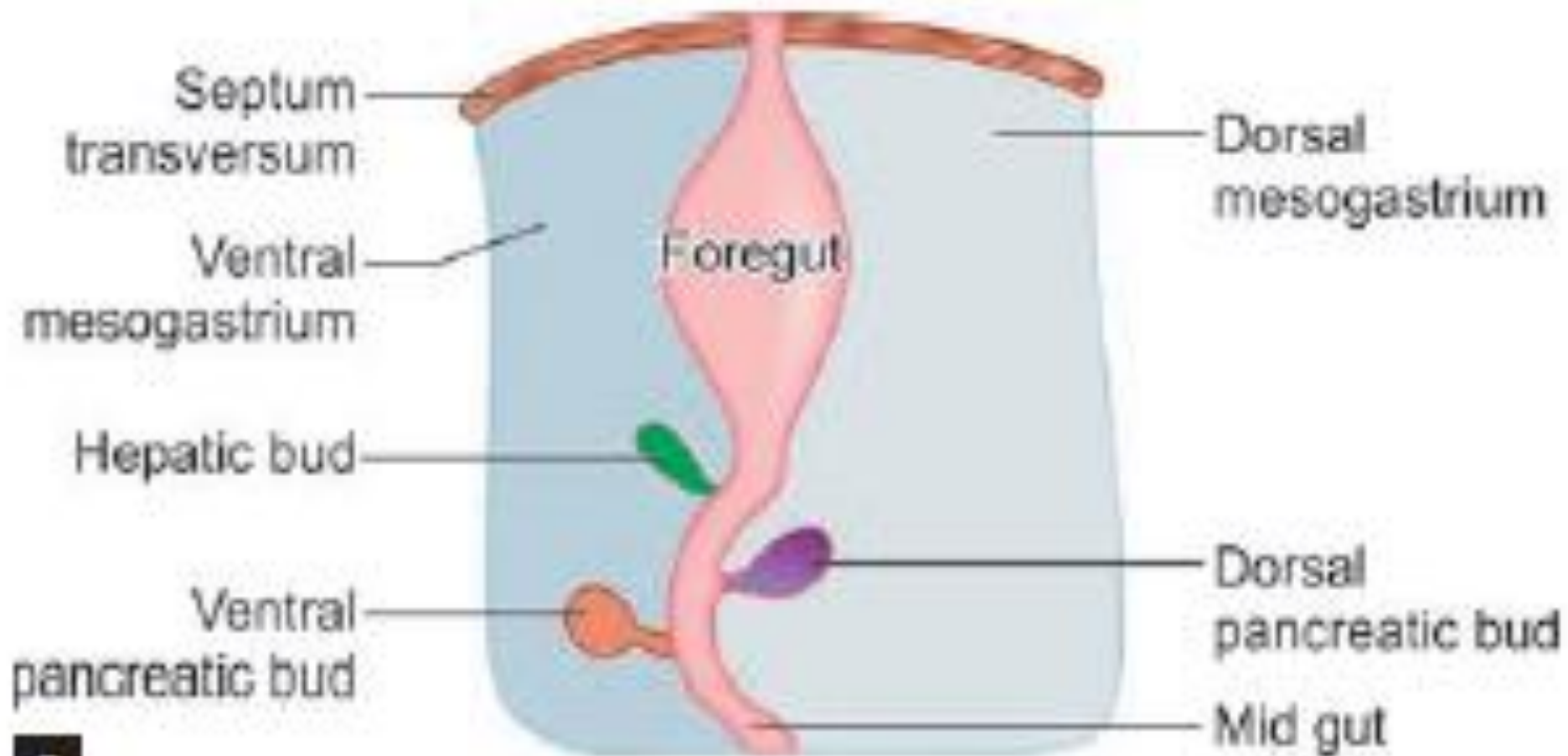
- derived from both foregut & midgut duodenum is supplied by branches of the celiac & superior mesenteric arteries that supply these parts of the primordial gut
- 5th & 6th weeks the lumen of the duodenum becomes progressively smaller
- temporarily obliterated by proliferation of its epithelial cells
- vacuolation occurs as the epithelial cells degenerate
- becomes recanalized by the end of the embryonic period
- most of the ventral mesentery of the duodenum has disappeared

- **Development of the Liver and Biliary Apparatus**
- ***Developmental primordium:***
- largest gland in the body
- functions include exocrine, endocrine, hematopoietic, metabolic & phagocytic
- develops from endodermal *hepatic bud* during 4th week of intrauterine life (IUL)
- arises from ventral margin of terminal part of foregut that forms upper half of 2nd part of duodenum

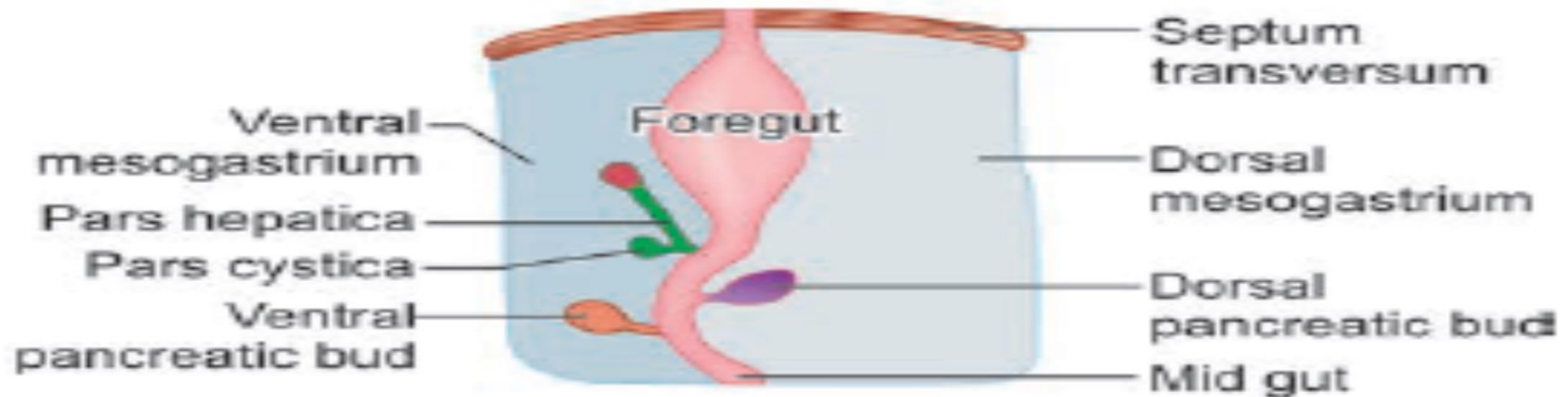




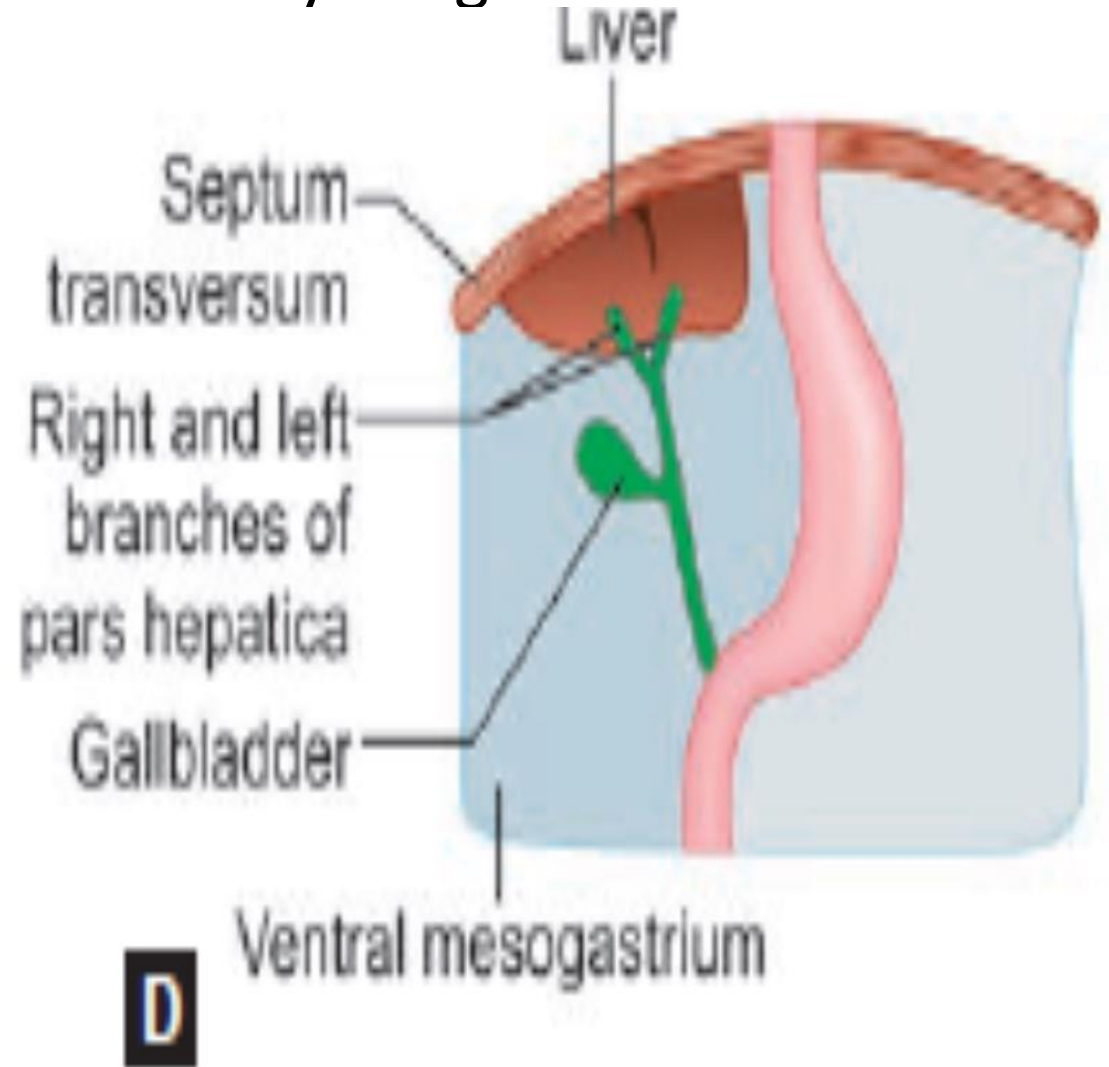
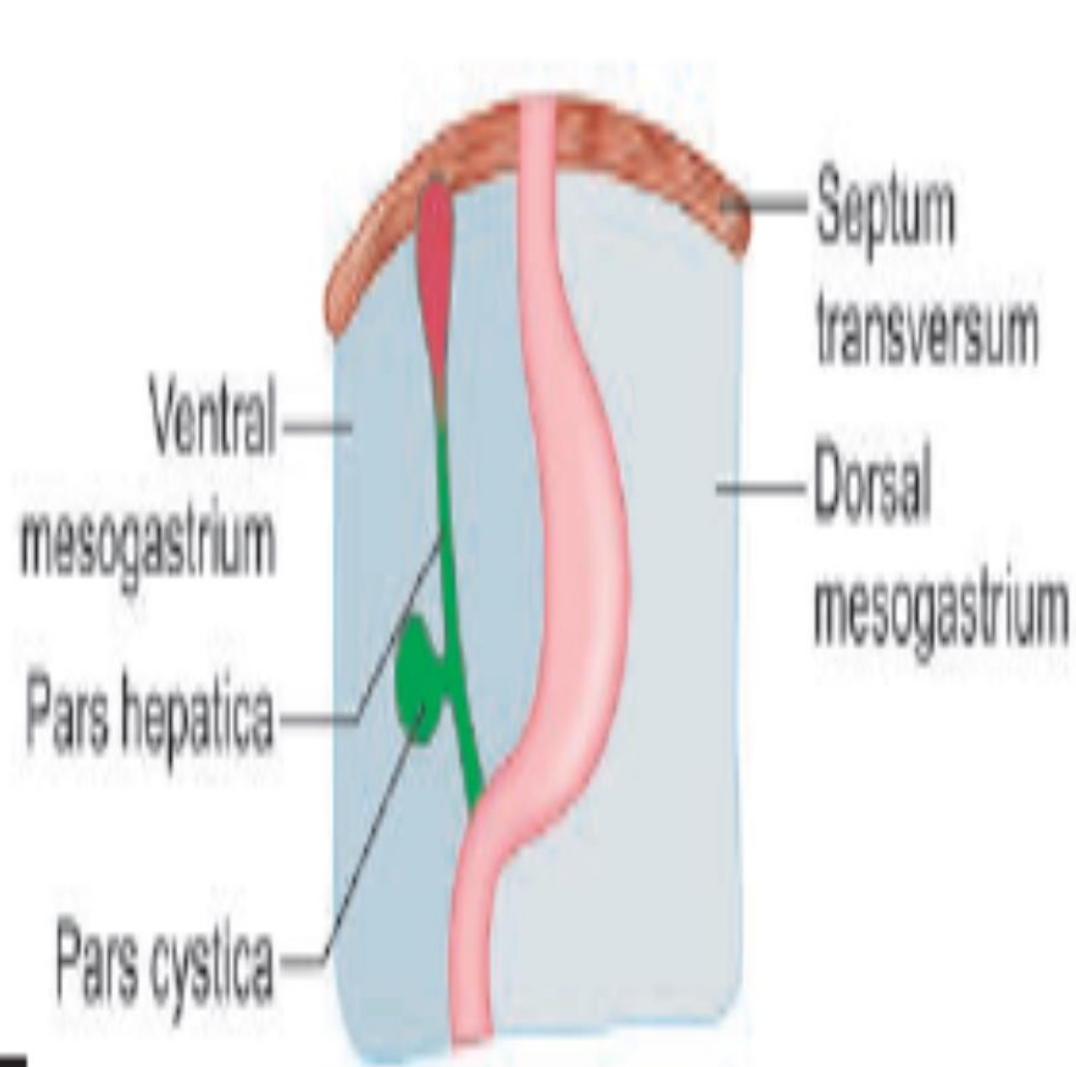
- Hepatic bud consists of proliferating endodermal cells that grow:-
- (a) ventrally & cranially into the *ventral mesogastrium*
- (b) through it into the ***septum transversum***
- **Septum transverse** (Unsplit part of intraembryonic mesoderm between pericardial cavity and yolk sac cavity)



- hepatic bud elongates & divides into :-
- cranial part ***pars hepatica*** that forms the liver
- caudal part ***pars cystica*** that forms gallbladder & cystic duct
- ***pars hepatica***: divides into right & left branches that become *right & left hepatic ducts*
- terminations of hepatic ducts contribute for formation of 2 solid *right & left lobes of liver*
- 2 lobes are of equal size during early development but size of left lobe reduces



- 3rd month of intrauterine life (IUL), the weight of liver is one-tenth of total body weight of fetus and occupies most of the upper abdomen
- 7th month of IUL it reduces to one-fifth of body weight



- ***Formation of hepatic architecture:***
- terminal part of right & left branches of pars hepatica (hepatic ducts), reach septum transversum
- clusters of cells (hepatocytes) in the form of laminae arise, break up into interlacing columns called ***hepatic trabeculae***
- between hepatic trabeculae the *hepatic sinusoids* develop in situ
- *vitelline & umbilical veins* running longitudinally in the septum transversum
- break up and establish communication with the hepatic sinusoids

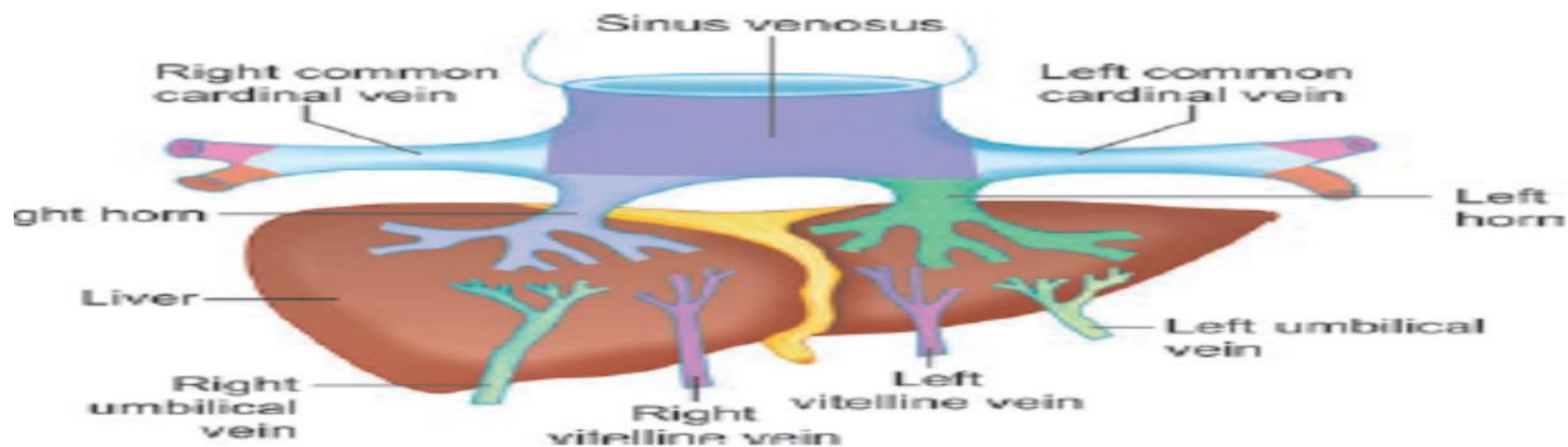
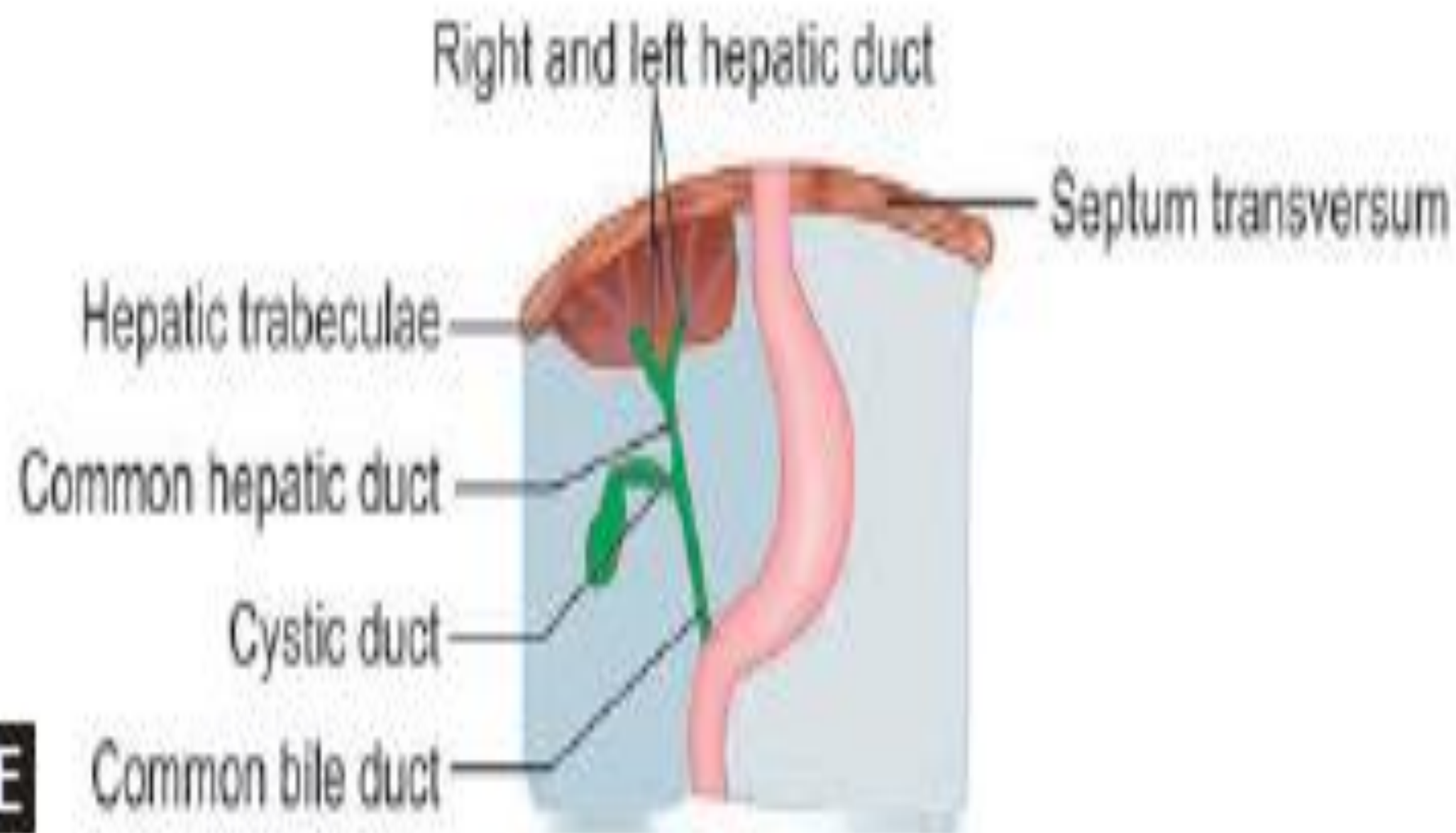


Fig. 14.4: Breaking up of umbilical and vitelline veins in septum transversum and their communication with hepatic sinusoids



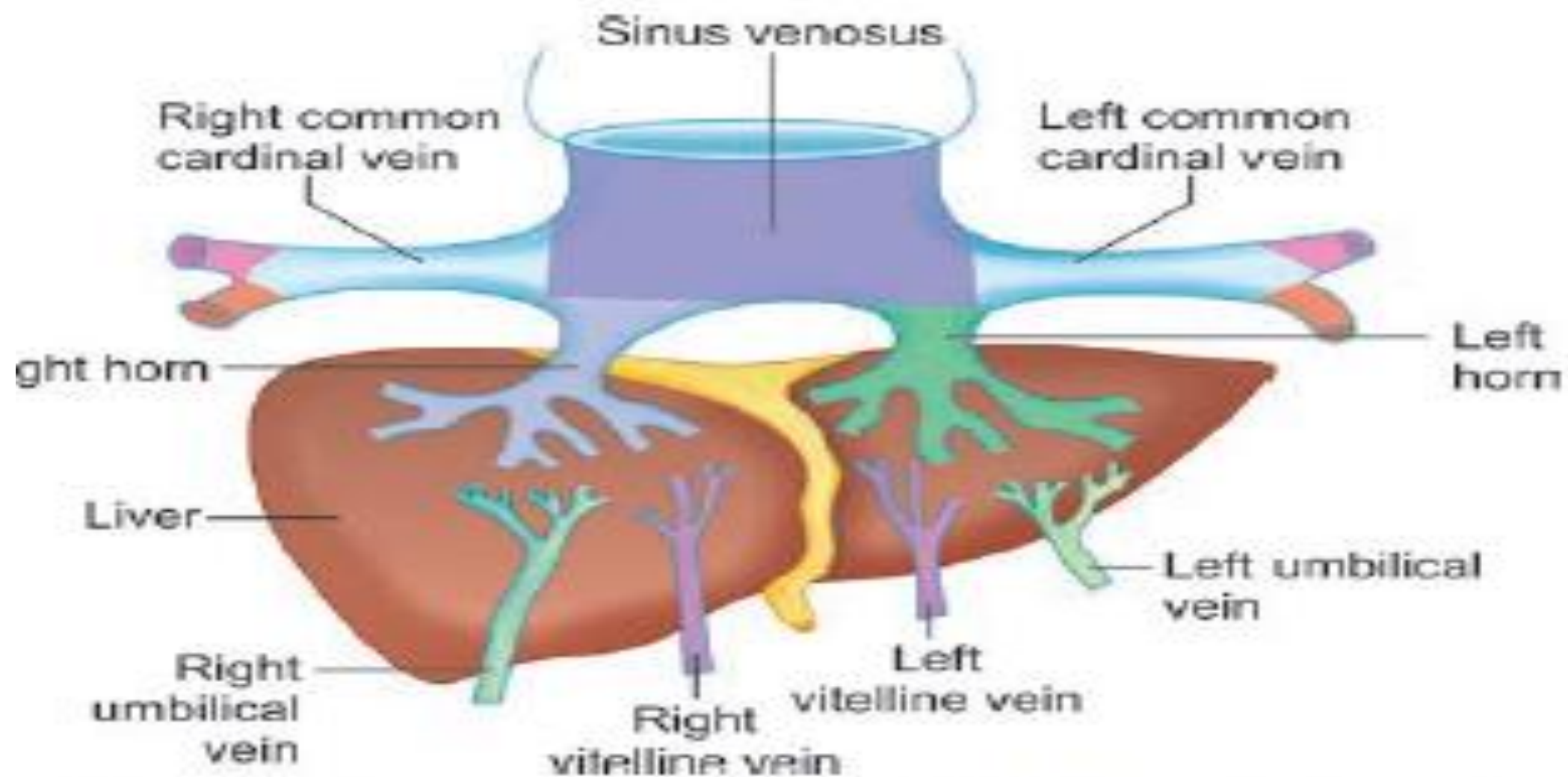
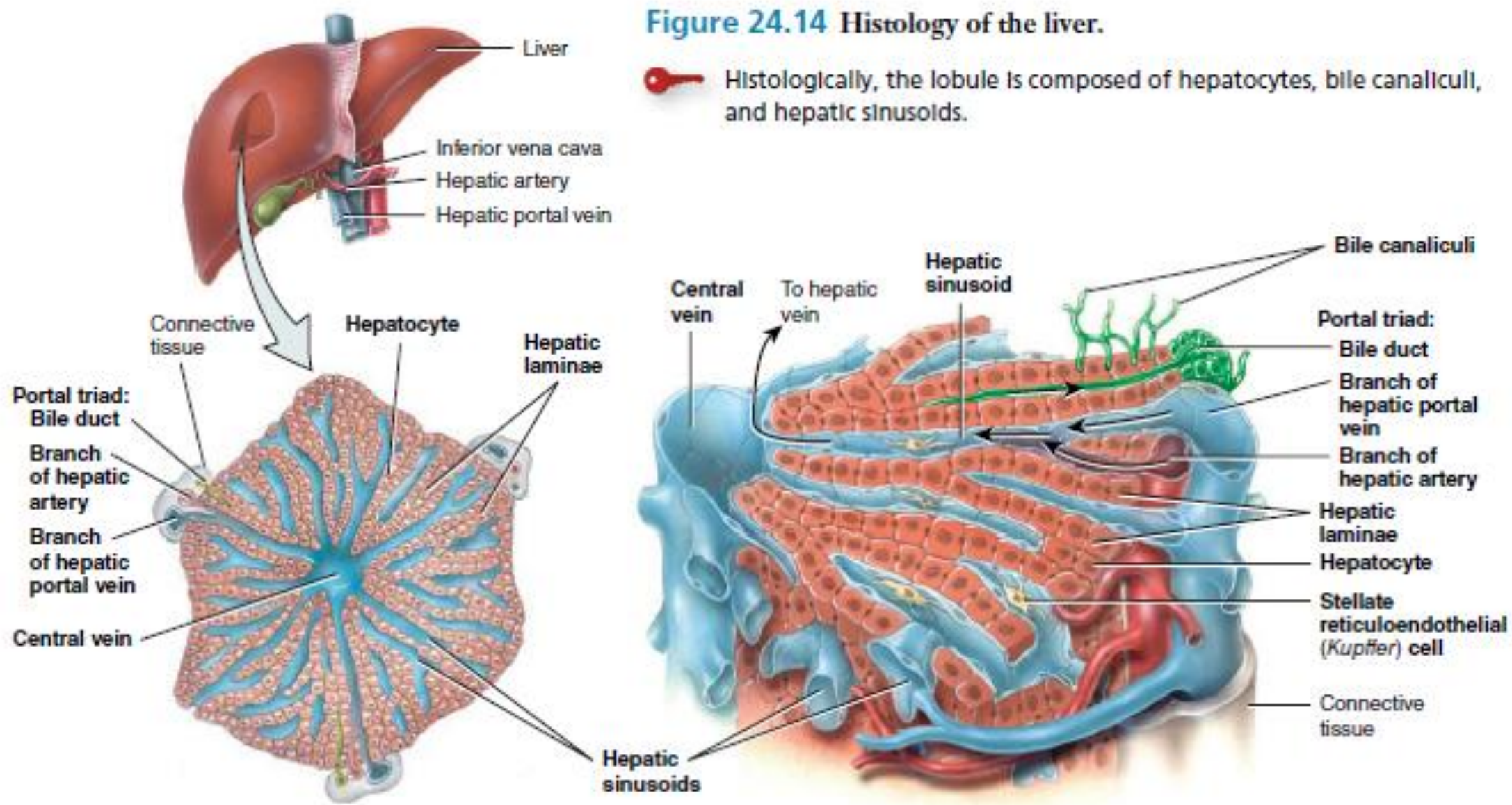


Fig. 14.4: Breaking up of umbilical and vitelline veins in septum transversum and their communication with hepatic sinusoids

- Within the substance of liver hepatic ducts branch repeatedly & canalized to acquire a lumen to form *intrahepatic biliary passages*
- hepatic trabeculae differentiate into the components of parenchyma, i.e. *liver cells* and *cells lining intrahepatic biliary system*
- Septum transversum contributes for the *Kupffer cells, hematopoietic cells & connective tissue cells*
- Reorganization of cells of hepatic bud & the mesenchymal cells of connective tissue and blood vessels form :-
- **A** -the hepatic lobule with bile canaliculi
- **B**- portal triad
- **C** -sinusoids of liver

Figure 24.14 Histology of the liver.

Histologically, the lobule is composed of hepatocytes, bile canaliculi, and hepatic sinusoids.

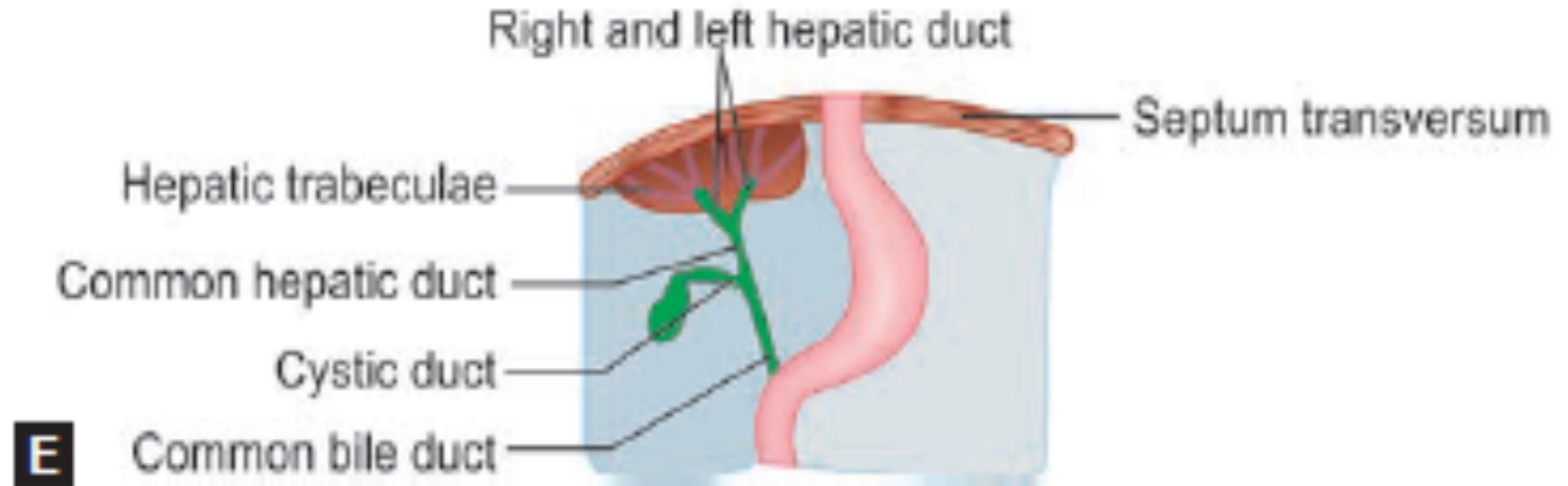


(a) Overview of histological components of liver

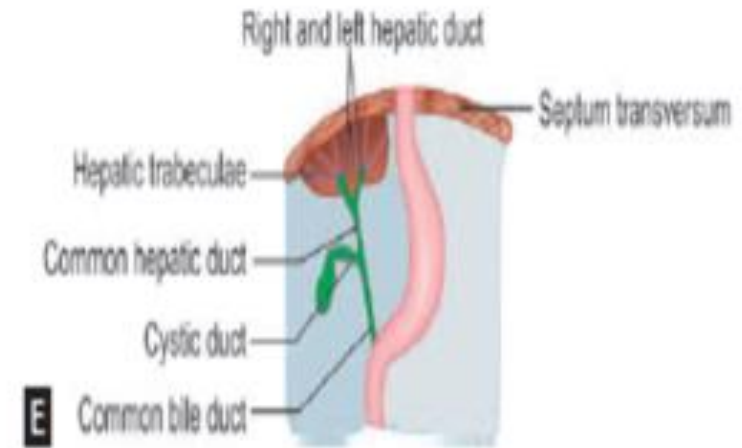
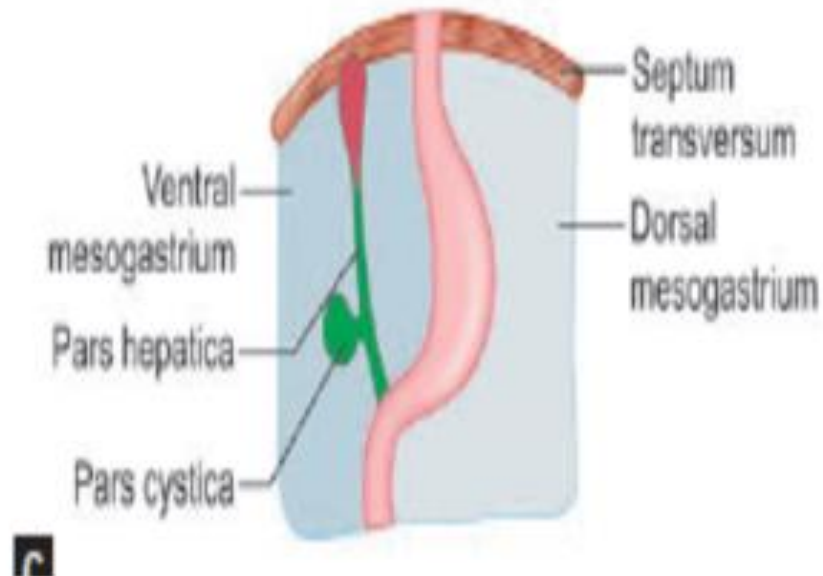
(b) Details of histological components of liver

- **Functions of Fetal Liver**
- ***Hematopoiesis***: begins 6th week of IUL & continues up to birth
- After birth function is carried out by spleen & bone marrow
- ***Bile secretion***: Secretion of bile by hepatocytes starts during 12th week of IUL
- released into the foregut derived part of duodenum
- then passes through the rest of intestine
- first stool passed by the new born is green in color due to the excretion of bile and called ***meconium***

- **Gallbladder and extra hepatic biliary passages (Extra hepatic biliary apparatus)**
- *Gallbladder* and *cystic duct* develop from pars cystica of hepatic bud
- narrow portion of hepatic bud between pars cystica & duodenal part of foregut forms *common bile duct*
- undivided part of pars hepatica distal to origin of pars cystica forms *common hepatic duct*
- right & left branches of pars hepatica become *right and left hepatic ducts*

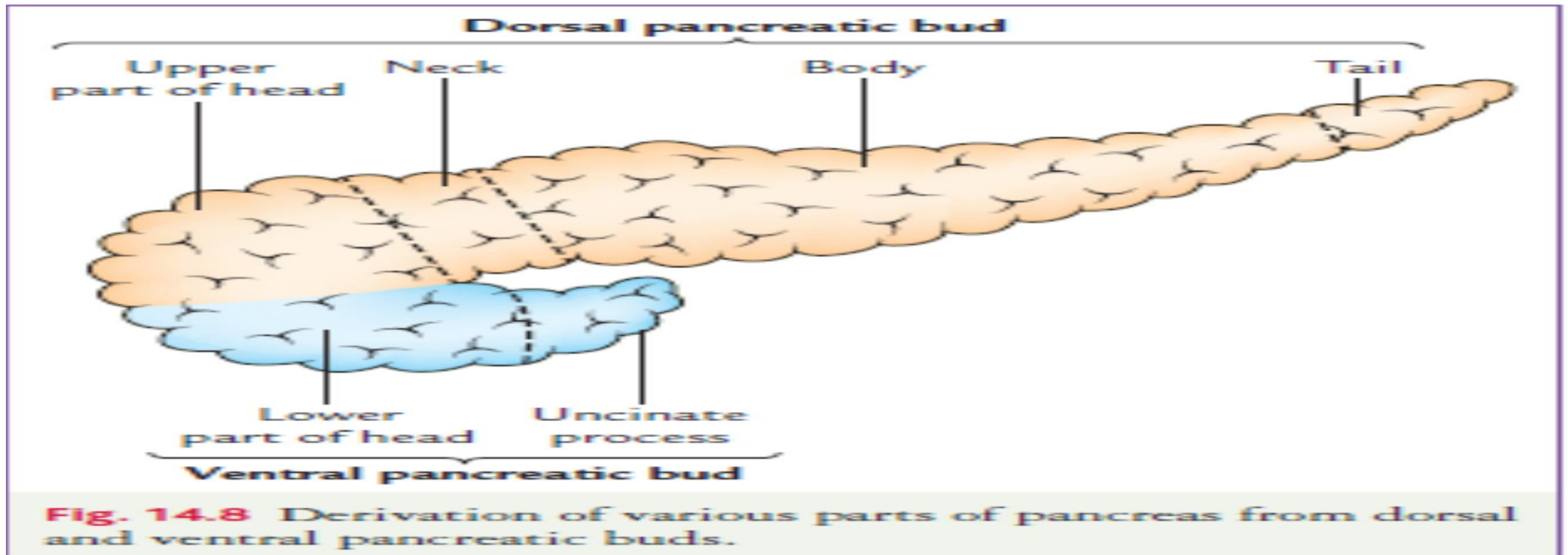


- bile duct at first opens on the ventral aspect of developing duodenum
- Due to the differential growth of duodenal wall, rotation of duodenal loop, it opens on the **dorsomedial** aspect of duodenum along with ventral pancreatic bud



- **PANCREAS**

- Functionally is both exocrine & endocrine gland with difference in microscopic structure
- developmental primordium of the 2 structural & functional components is common
- develops from 2 endodermal buds, *dorsal & ventral pancreatic buds*

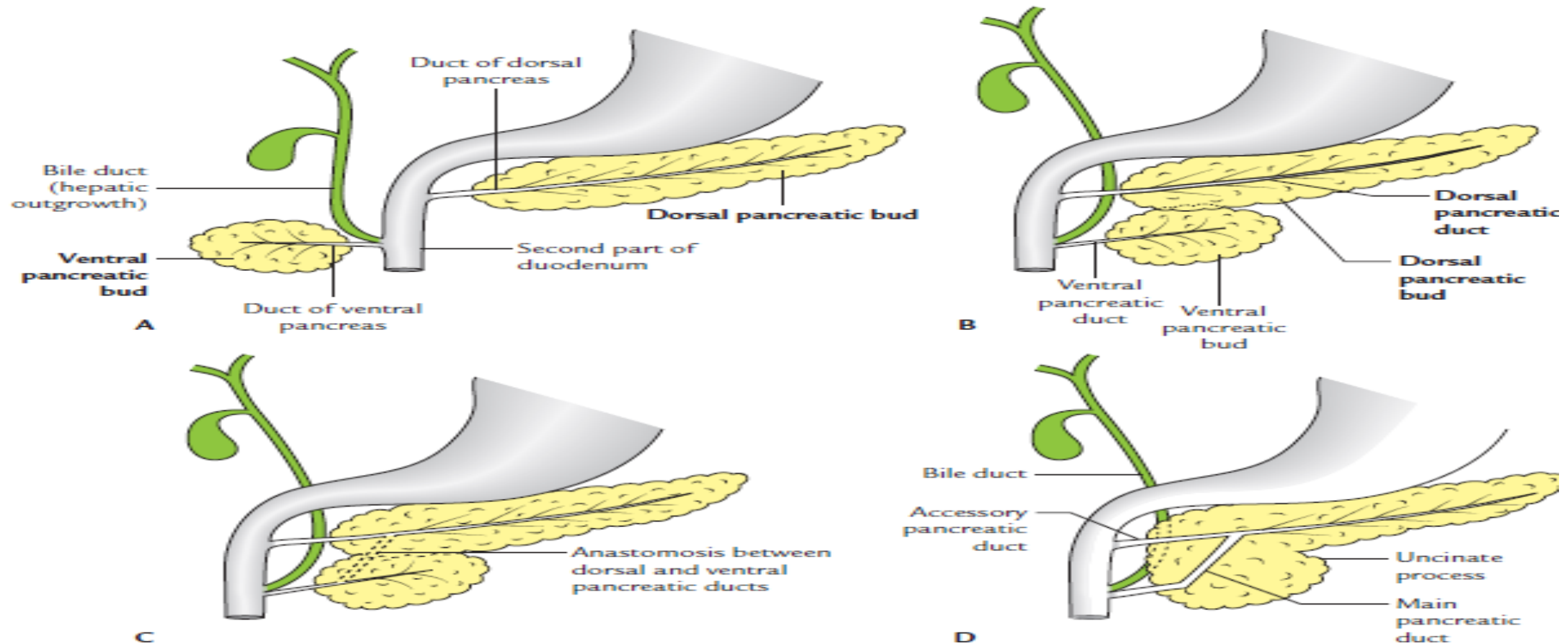


- dorsal & ventral pancreatic buds arise from :-
- ❖ dorsal & ventral walls of terminal part of foregut (future second part of duodenum)
- ❖ caudal to hepatic bud before rotation of midgut

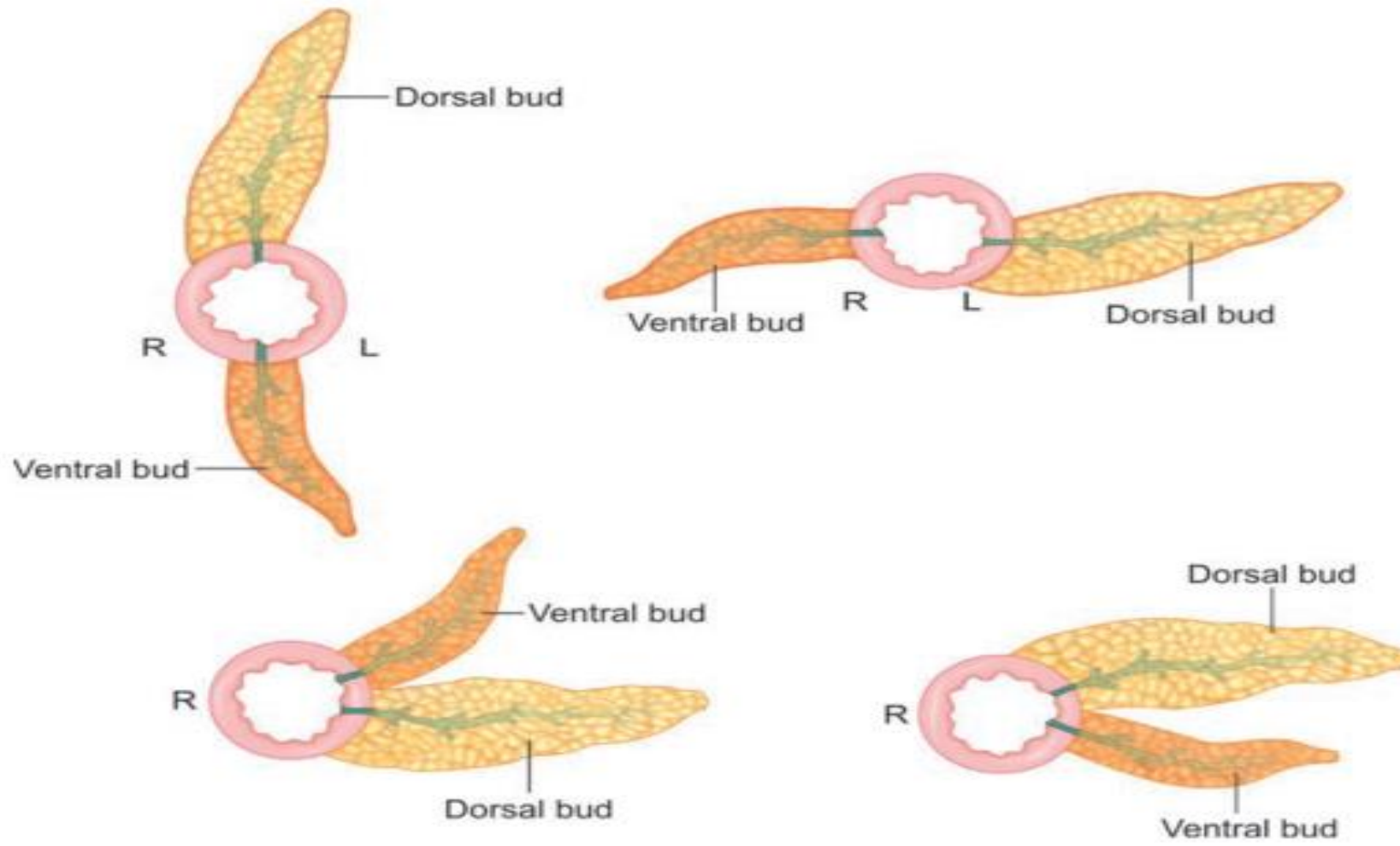
Dorsal pancreatic bud:

- First to appear (4th week) & larger in size.
- cephalic to the ventral bud.
- grows between 2 layers of dorsal mesentery of duodenum

- ***Ventral pancreatic bud:***
- arises in close relation to hepatic bud
- appears later & smaller than dorsal bud
- grows between the 2 layers of ventral mesentery of duodenum



- ***Change in the position of pancreatic buds:***
- the rotation of duodenal loop to the right
- ventral pancreatic bud along with primitive bile duct comes to the right
- the dorsal bud comes to the left of duodenum



- differential growth of wall of gut
- attachment of ventral pancreatic bud along with primitive bile duct (hepatic bud derivative) shifts to the left moving closer to the dorsal pancreatic bud
- Pancreatic tissue formed from ventral & dorsal pancreatic buds fuse to form one mass in 7th week of IUL
- Ventral pancreatic bud forms lower part of head & uncinate process of pancreas
- Dorsal pancreatic bud forms upper part of head, neck, body & tail of pancreas

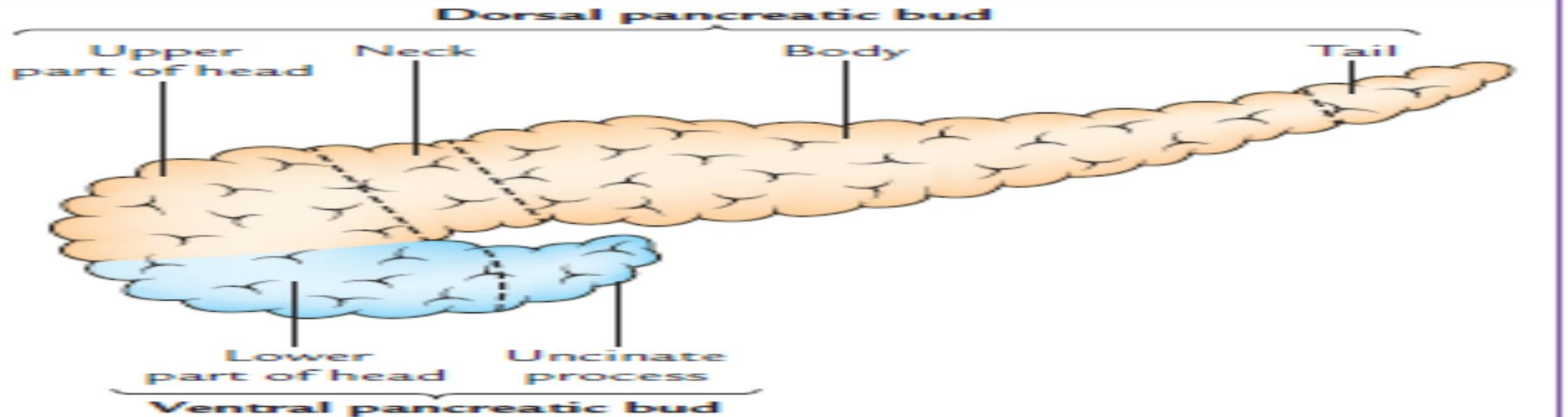
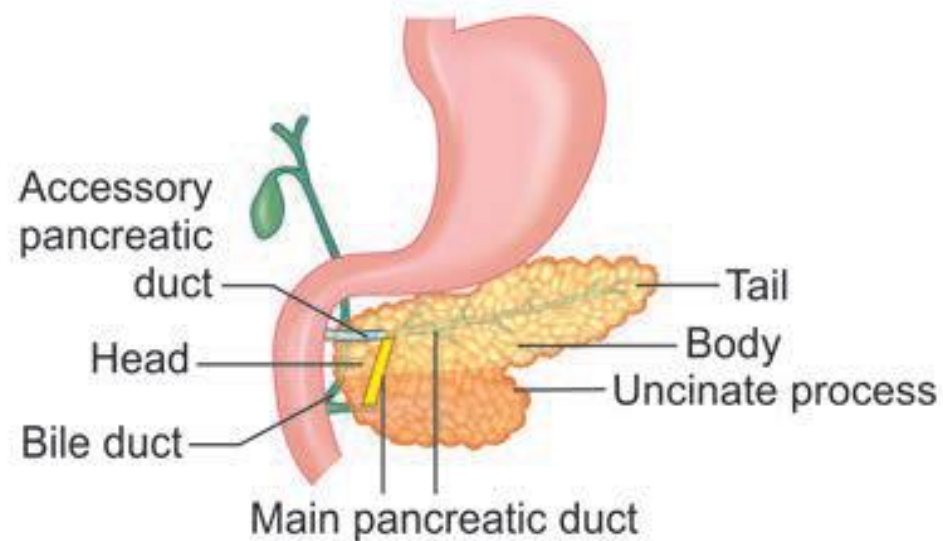
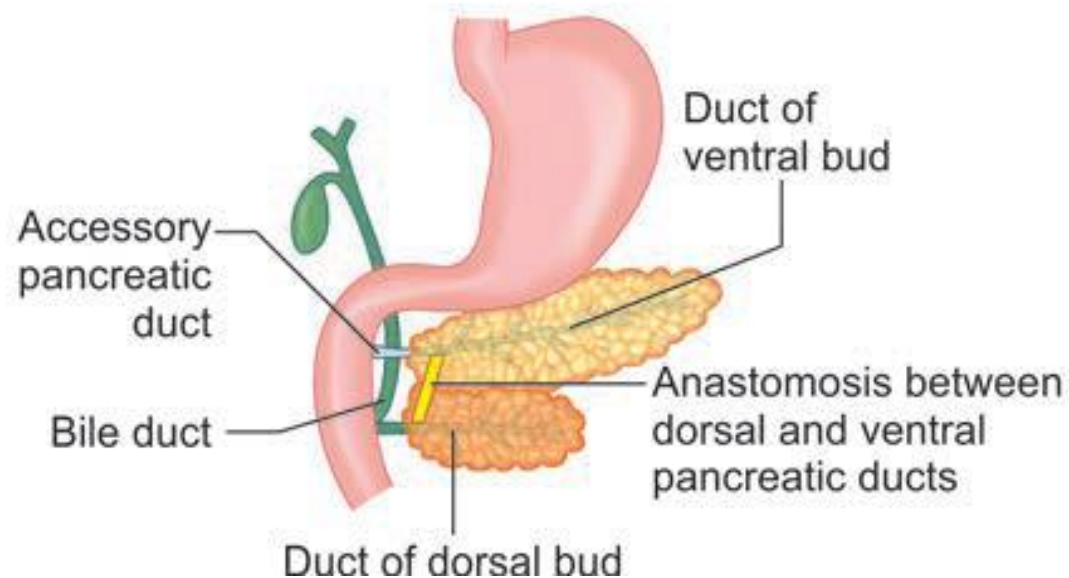
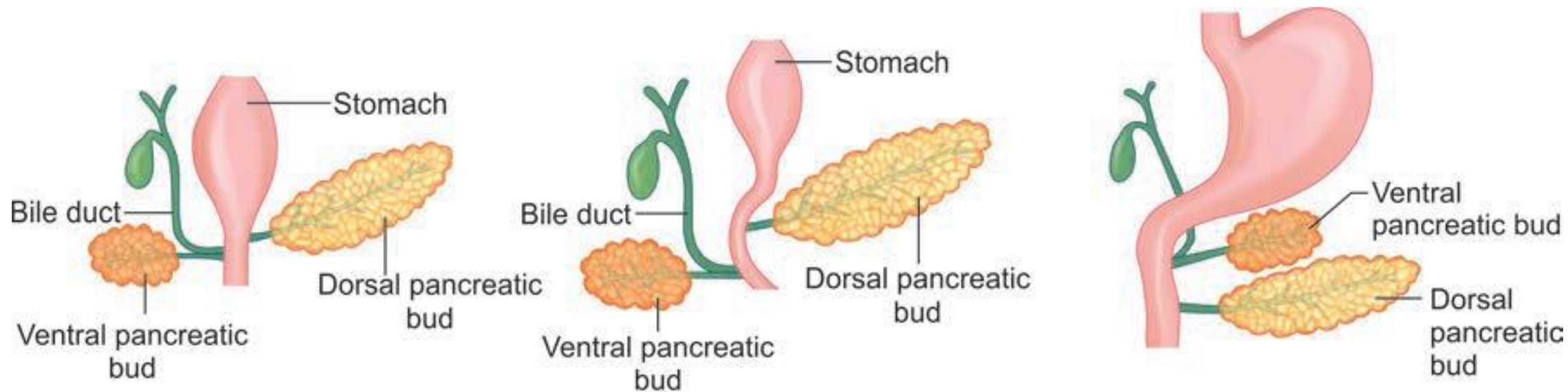
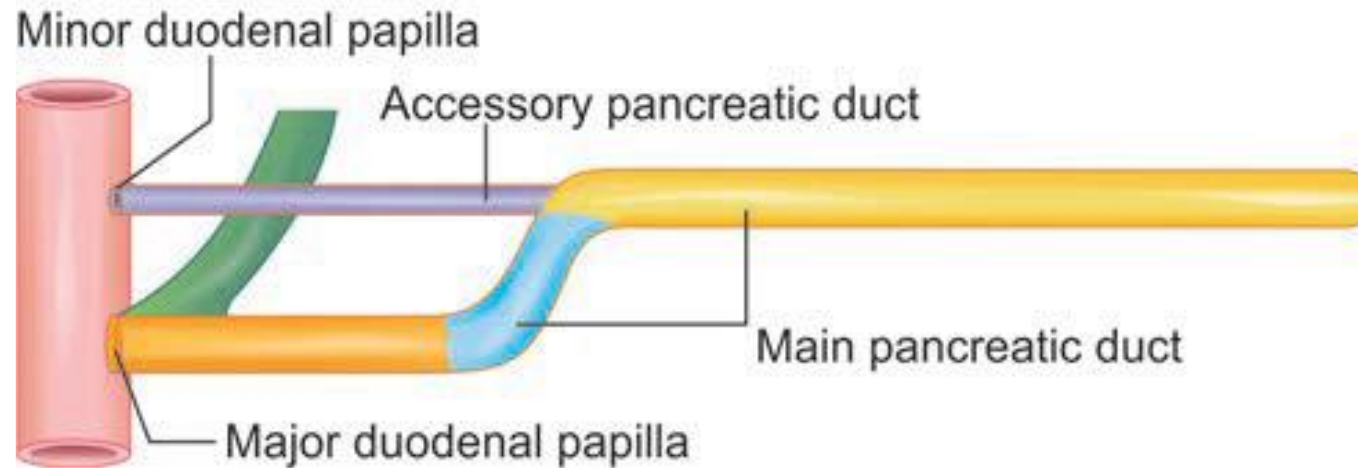
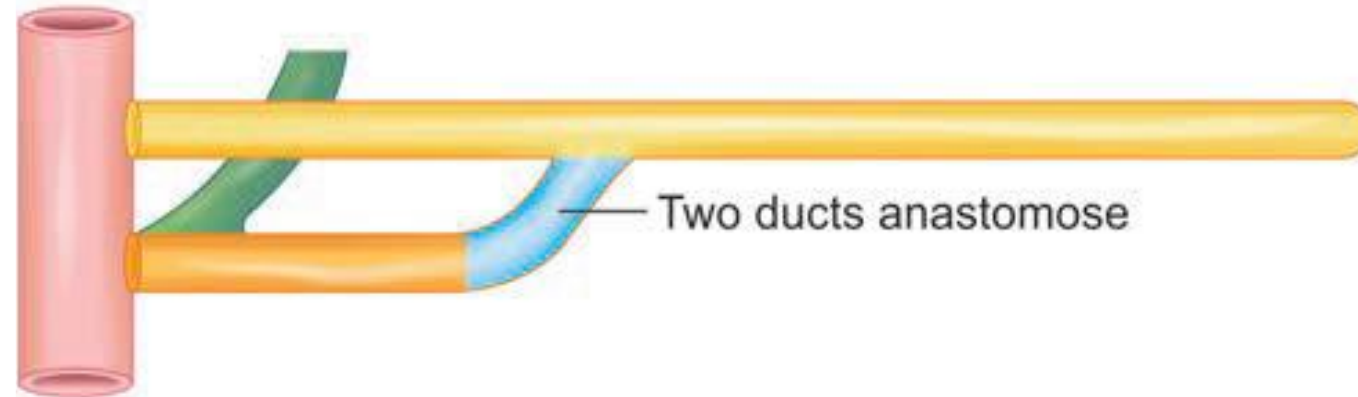
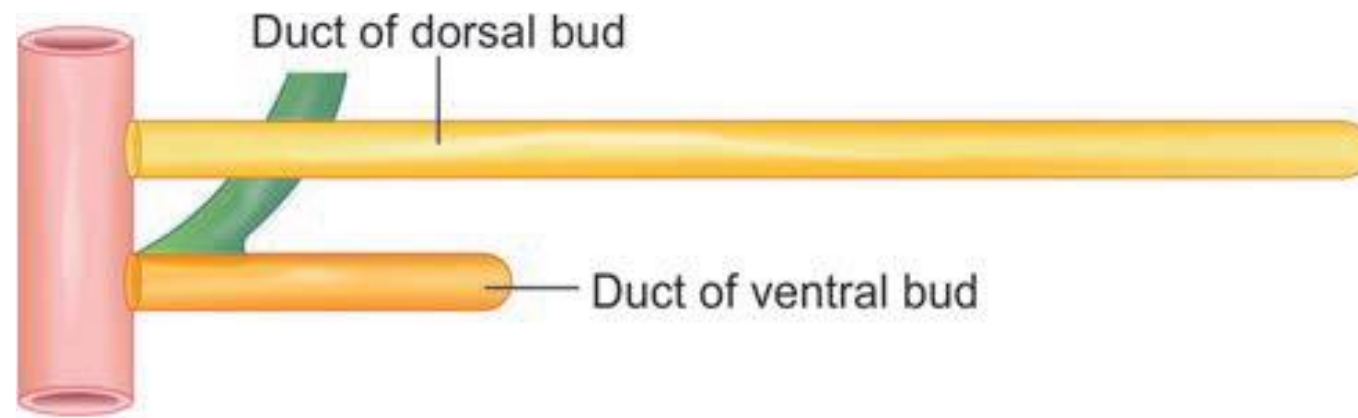


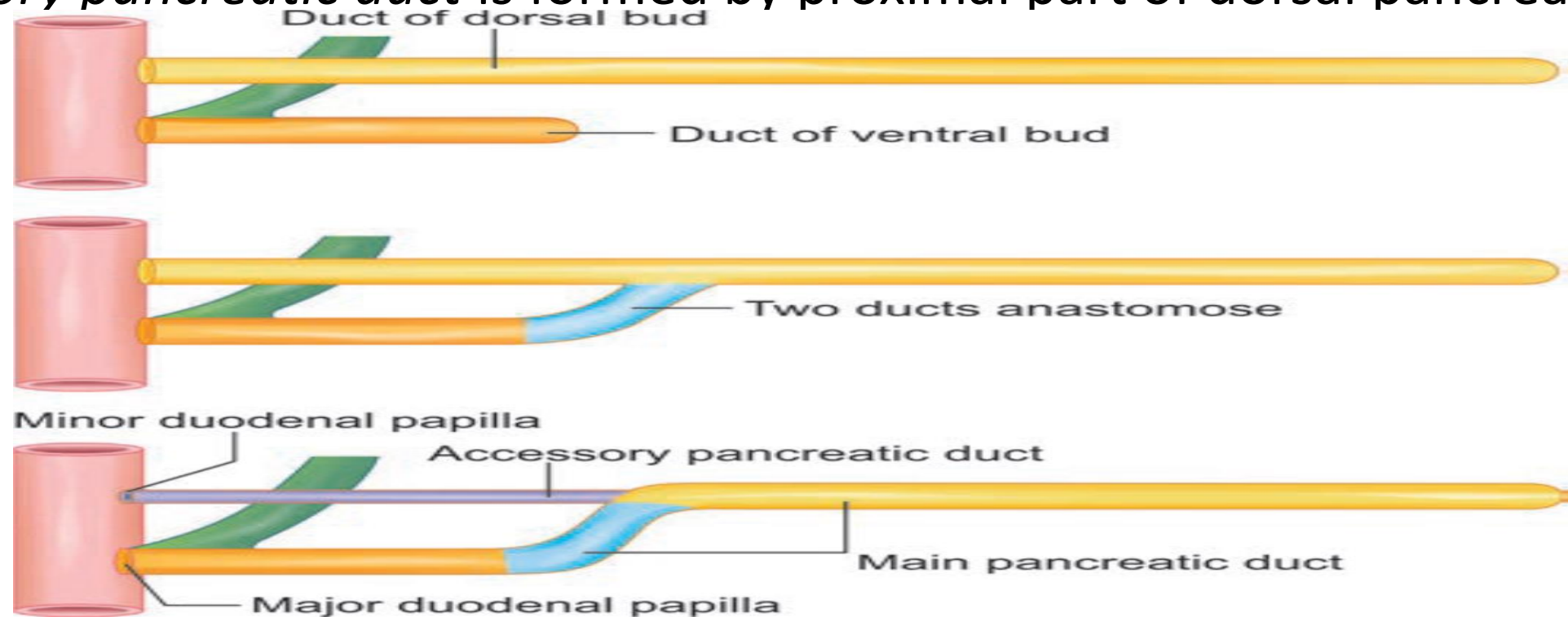
Fig. 14.8 Derivation of various parts of pancreas from dorsal and ventral pancreatic buds.

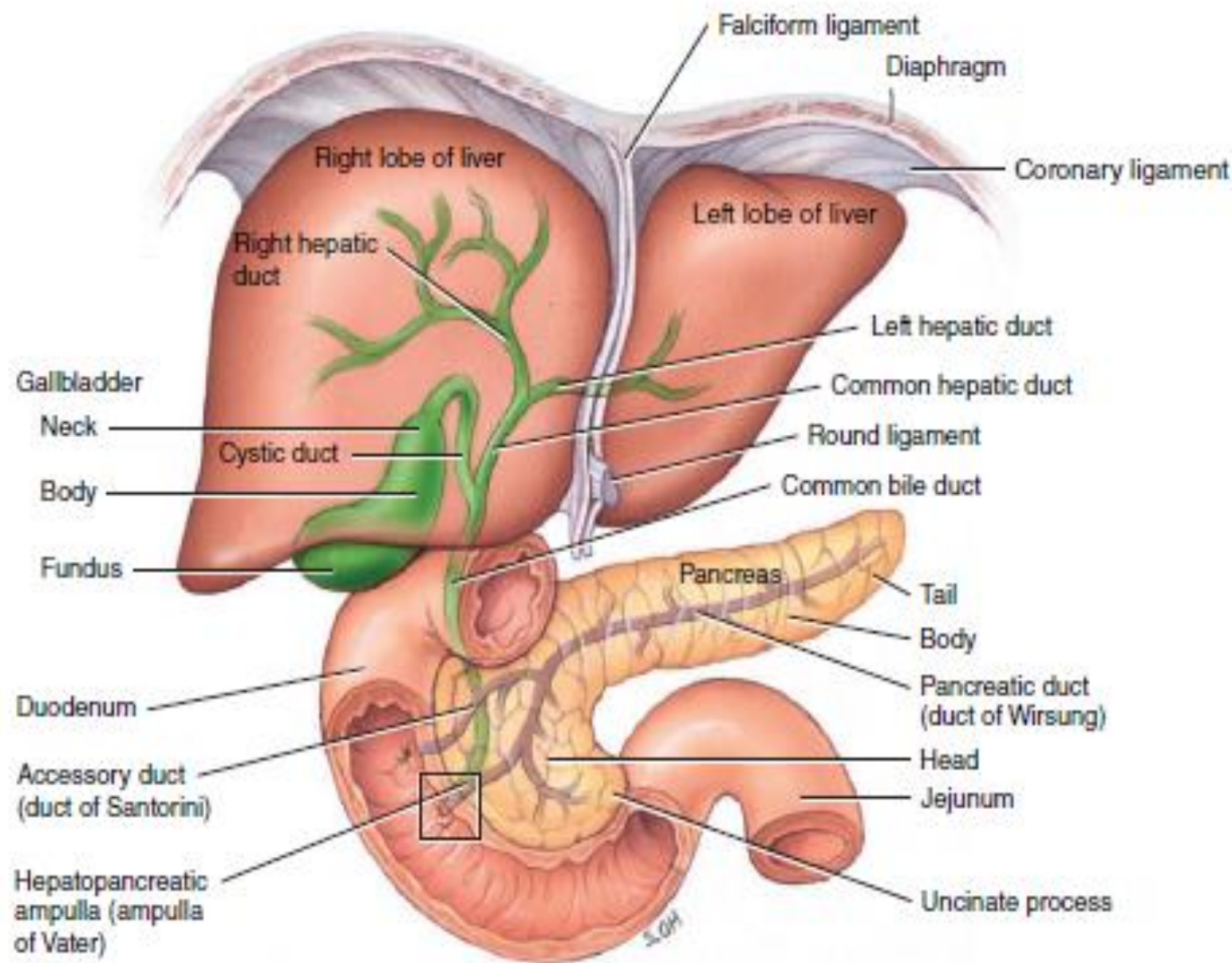


- ***Duct system of pancreas:***
- Initially ducts of dorsal & ventral pancreatic buds are separate & open separately into the duodenum
- Opening of dorsal pancreatic duct is 2.0 cm proximal to the opening of ventral pancreatic duct
- Ventral pancreatic duct & bile duct (hepatic bud derivative) have a common opening in the duodenum
- ducts of dorsal & ventral pancreatic buds anastomose establishing communication between the 2

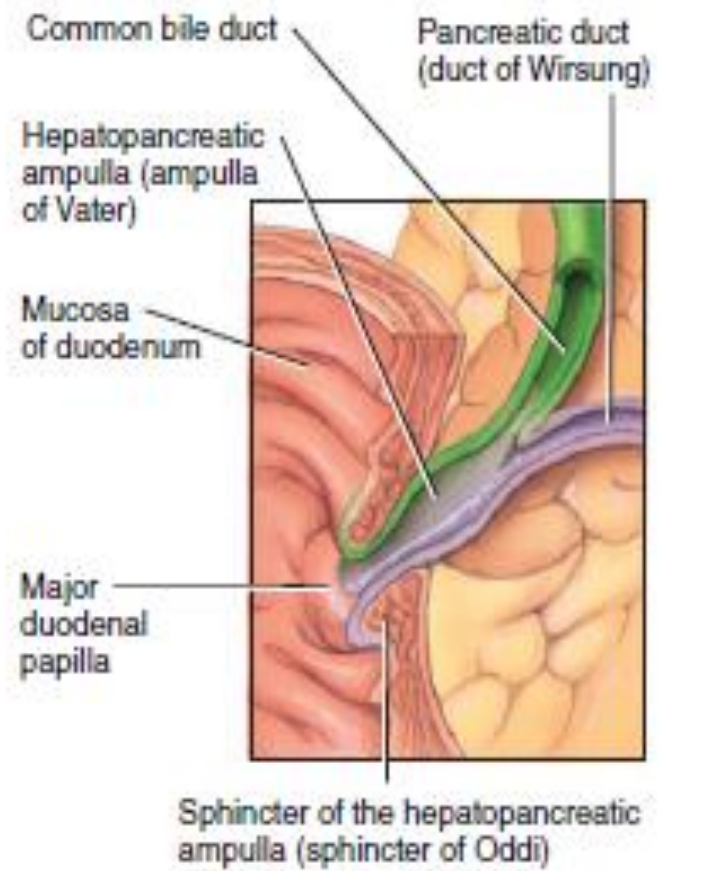


- ***main pancreatic duct*** is formed:-
- distal part, by the duct of dorsal bud,
- middle part by the oblique cross-communication between ducts of 2 buds
- proximal part by the duct of ventral bud
- main pancreatic duct opens into duodenum at major duodenal papilla, along with bile duct
- ***accessory pancreatic duct*** is formed by proximal part of dorsal pancreatic duct



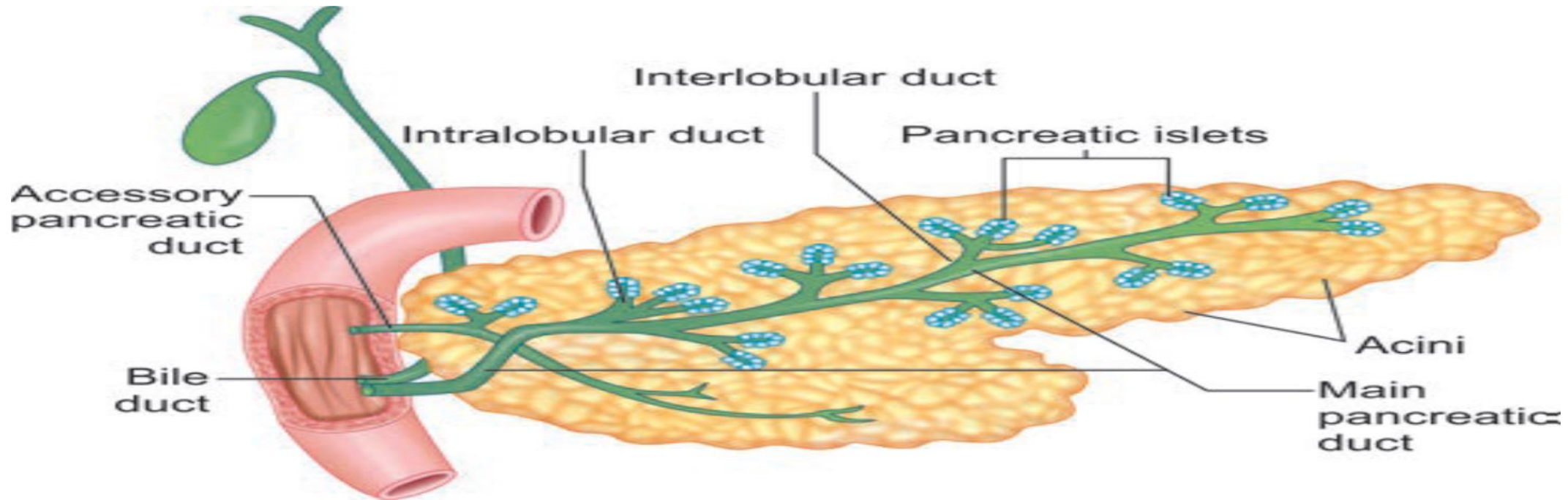


(a) Anterior view



(b) Details of hepatopancreatic ampulla

- Repeated branching of the major & minor pancreatic ducts forms the interlobular and intralobular ducts and ductules
- ***Parenchyma:***
- develops from branching of endodermal pancreatic buds into the surrounding mesoderm
- parenchyma consists of exocrine and endocrine secreting units
- exocrine part consists of acinar secreting units, develops from proliferation and reorganization of cells at the terminations (ductules) of duct system



- endocrine part, i.e. Islets of Langerhan's, develops from separation of groups of cells from the terminations of duct system
- ***Retroperitoneal location of entire pancreas except tail:***
- initially both buds were suspended in the respective mesogastria,
- due to their migration they occupy a position posterior to the peritoneum
- except the tail of pancreas which lies in the lienorenal ligament

- **Functions of Fetal Pancreas**

- Its exocrine function begins after birth
- Endocrine function begins in fetal and in early embryonic period.
- By 7th week α -cells start secreting glucagon
- by 10th week insulin production by b-cells begins

- **SPLEEN**
- lymphoid organ
- develops from mesoderm in the dorsal mesogastrium, close to the developing stomach
- **Spleniculi:**
- It develops as a collection of mesenchymal cells to form small lobular masses of splenic tissue (Spleniculi) in the dorsal mesogastrium
- lobules later fuse to form single mass of spleen
- Presence of splenic notches along the upper border of adult spleen indicates lobulated development of spleen



Fig. 14.25: Multiple notches along superior border of spleen indicating persistence of fetal lobulation

- splenic projection in dorsal mesogastrium divides into:-
 - ❖ an anterior part extending from the stomach to the spleen the gastrosplenic ligament
 - ❖ a posterior part that extends from the spleen to the posterior abdominal wall the lienorenal ligament
- posterior layer of dorsal mesogastrium fuses with the posterior abdominal wall

- It extends between spleen and left kidney forming the lienorenal ligament
- As a consequence of this fusion and change in orientation of stomach, the spleen comes to lie on the left side and takes part in forming left boundary of the lesser sac of peritoneum

- Histogenesis of Spleen
- All elements of the spleen are derived from mesoderm
- mesodermal cells form capsule, septa, and connective tissue network including reticular fibers
- primordium of splenic tissue forms branching cords and isolated free cell
- Some of the free cells form lymphoblasts while the others differentiate into hemopoietic cells
- process of blood formation in spleen begins in early embryonic life and continues during fetal life but stops after birth.
- The production of lymphocytes, continues in the postnatal period

• **Clinical Application**

- Esophageal Atresia
- Esophageal Stenosis
- Tracheoesophageal fistula
- Achalasia cardia:
- Short Esophagus (Congenital Hiatal Hernia)
- Congenital Hypertrophic Pyloric Stenosis
- Duodenal Stenosis
- Duodenal Atresia
- Anomalies of the Liver
- Accessory Pancreatic Tissue
- Anular Pancreas

- Accessory spleen (spleniculi)
- Lobulated spleen
- Anomalies of gallbladder